# Game Engine Usability: Unity vs. Godot

## A Usability Study of the Unity and Godot Game Engines

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By Connor Wolfe

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# 1 | Introduction

Game developers, especially new game developers, need to make a conscious decision on which game engine to use. Unity, Unreal, and Godot are the currently most widely adopted game engines. Unreal is highly specialized for graphically intense 3d games and so the decision to use that editor is decided when a game developer decides to make a game in that scope. Unity is a more versatile engine and is the most widely adopted game engine. However, Godot has been on the rise as a direct competitor to Unity; as a free, open source, license free alternative. That said, a game engine that is hard to use will impact game developers and their work, so it is important to know how these engines' usability stacks up when making this important decision.

So, in this white paper these two engines (Unity and Godot) will be compared on their usability using criteria and methodology that is explained in the following sections. The 10 tasks defined are designed to encompass the tasks that the average game developer will need to do within their engine on a regular basis. By the end of this paper, game developers will be able to make a more conscious decision on which game engine to use.

# 2 | Methodology

## 2.1 | Environment

This testing was done in an apartment room. However, these tests can be done anywhere with internet access. The tasks were completed with few distractions, closed doors, and complete focus.

### 2.1.1 | Prerequisites

The Unity Hub (and Editor) and Godot Editor must be installed from their websites and then opened. For the Unity Editor you must also have an external Code Editor, like Visual Studio, to write and edit code/scripts in the editor.

### 2.1.2 | Equipment

The tests in this study were ran on a Windows 11 desktop running Unity Hub 3.15.1.0, Unity Editor 6000.2.3f1, and Godot Editor v4.5.1-stable_win64.

## 2.2 | Tasks

1. Create a new project
2. Create a script
3. Save Project Files
4. Determine the cause of a simple bug
5. Run the scene/project

6. Import an Asset
7. Change the Layout
8. Import an Existing Project
9. Install an Extension
10. Export/Build the Project

## 2.3 | Criteria

Each test will be compared on the following 5 metrics.

- **Effective**: Are users successful in completing the task within the product
- **Efficient**: Can users do the task with the lowest amount of time or effort
- **Engaging**: Is the product pleasing to use
- **Error Tolerant**: How well does it assist in handling/recovering from errors
- **Easy to Learn**: How intuitive is it to learn to use the product

## 2.4 | Rating Table

For each of the Criteria listed above I will be rating each task on a 6-point (Strongly disagree to Strongly agree) Likert Table. The following is the 6-point Likert table.

| Strongly Disagree | Disagree | Slightly Disagree | Slightly Agree | Agree | Strongly Agree |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |

# 3 | Task Results

## 3.1 | Create a new project

### 3.1.1 | Unity

After selecting "+ New Project", Unity presents templates for you to start your project with, as seen in figure 2 Unity uses templates to give a baseline when starting project. Its mostly a fake choice but its relatively intuitive and it does help streamline development.

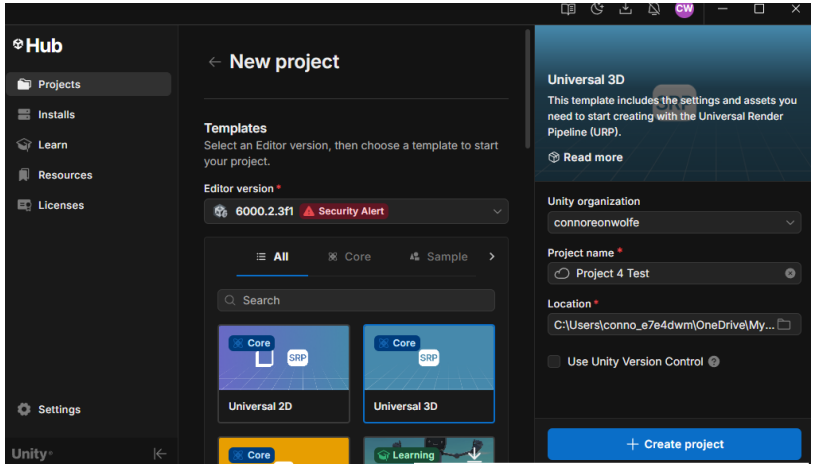



Figure 2: Unity Create menu

### 3.1.2 | Godot

Godot makes things a bit easier than Unity, just 2 clicks and your in your new project. Everything is nicely labeled, and is very intuitive. See figure 3 for Godot's create menu.

So, both engines scored very highly for this task across the board as they were both low effort and efficient. Making sure starting the project is not a barrier to entry is super important, so it is good that both engines scored highly on this task.

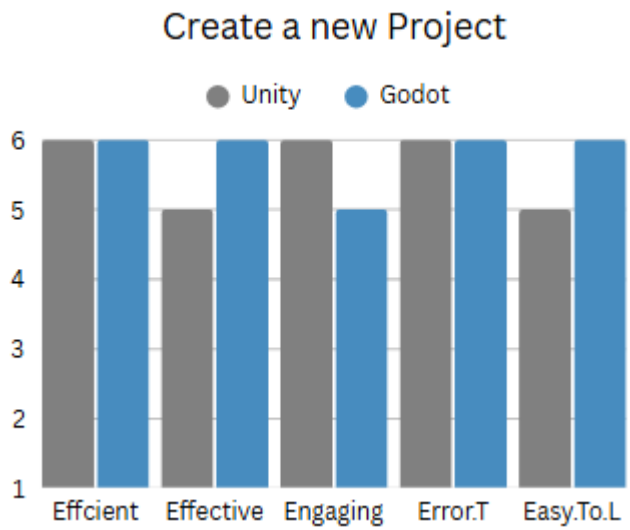



Figure 1: task 1 results bar graph

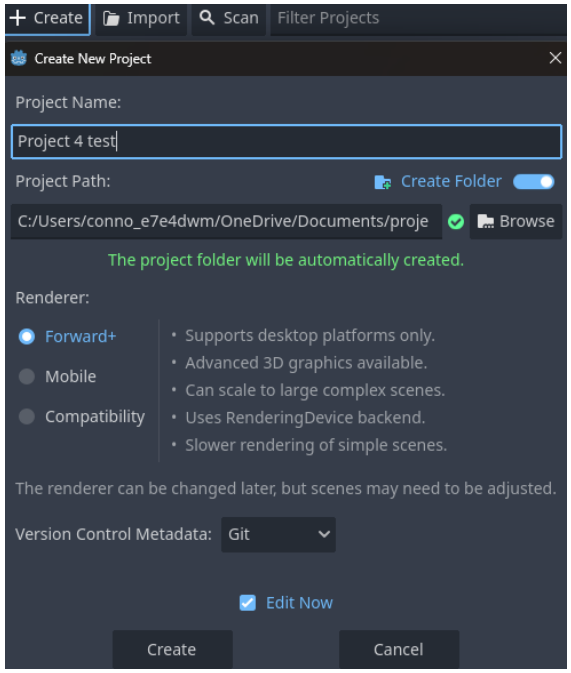



Figure 3: Godot Create menu

## 3.2 | Create a script

### 3.2.1 | Unity

It's super easy to make a script in both engines, but Unity does have it a couple clicks in from a right click. Which does mean it takes a lot more than 2 clicks to get the script, which is why Godot scored a little higher on Effectiveness and Easy to Learn. See figure 5 for the click path it takes to create a script.

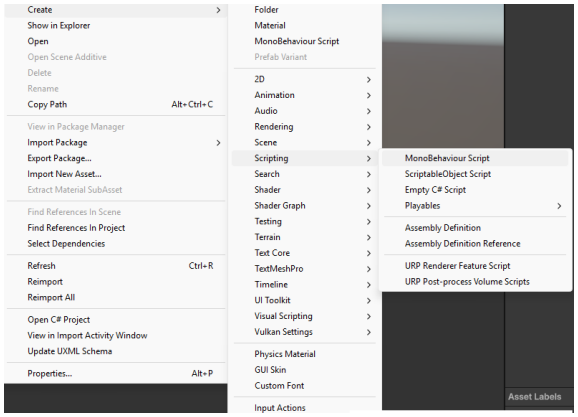



Figure 5: Unity

### 3.2.2 | Godot

Making a script in Godot only took 2 clicks and I had a script. So super easy and quick. The only thing is that Godot's error message for invalid pathing is not very clear, so it scored a little lower on Error Tolerant, see figure 6 for this error.

So both engines scored very highly overall in this task across the board. For both engines it just takes a right click and one or two clicks to start coding in a script. Which is super important as coding/scripting is the most constantly repeated activity done in Game Development.

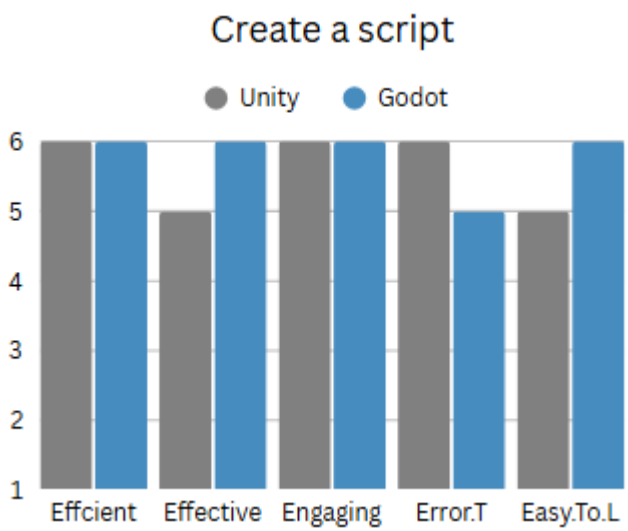



Figure 4: task 2 results bar graph

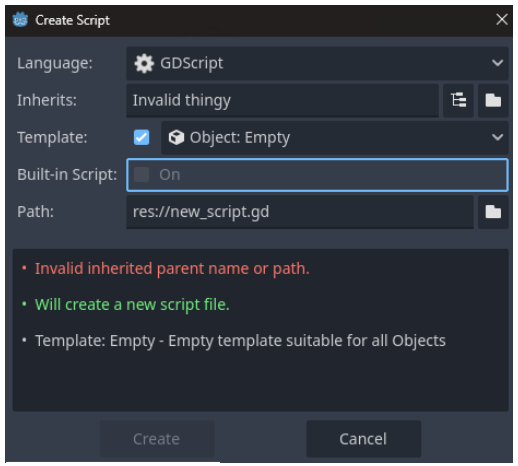



Figure 6: Godot

## 3.3 | Save the Project Files

### 3.3.1 | Unity

Unity scored perfectly across the board for this task. You can either use ctrl+S, or your can go to “File”->”Save”, or “File”->”save As...”. Which are all super intuitive and common for almost any editing software, not just necessarily software editor’s.

### 3.3.2 | Godot

Saving in Godot was very confusing from a first time user’s experience. This is as Godot has both a Scene and a Project drop down. And the save options are not in the project drop down but instead in the scene drop down, as seen in figure 8, which felt very unintuitive as a first time user. Luckly, Godot does also support ctrl+S for saving.

So for this task Godot scored poorly, as having to learn how to save your project every time you pick up the engine will be time consuming and frustrating.

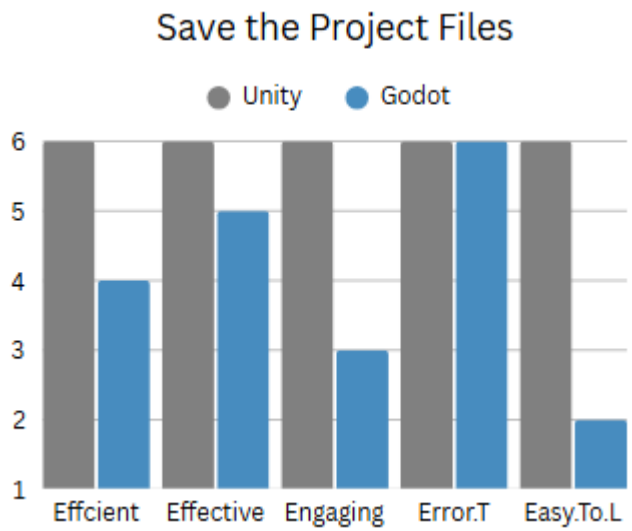



Figure 7: task 3 results bar graph

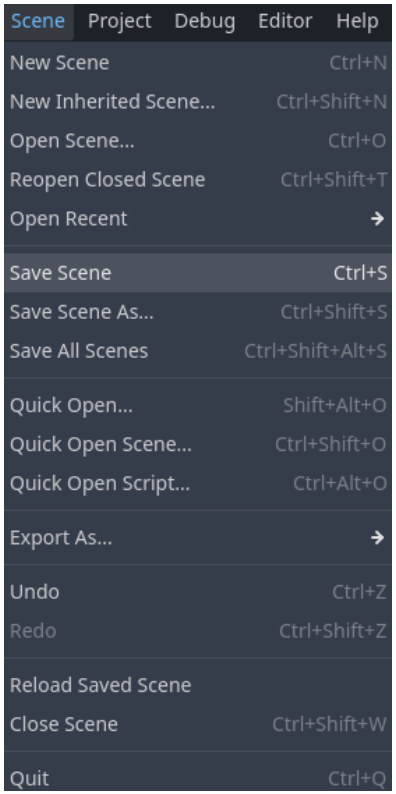



Figure 8: Godot

## 3.4 | Determine the cause of a simple bug

### 3.4.1 | Unity

Unity scored relatively poorly as Unity does not have a native editor. This can be a problem as that mean Unity can't determine specifically what is wrong with your code. For example, I did this test by writing the line "bogus line of code" and Unity could only make a vague guess as too what is wrong with the code, as seen in figure 10.

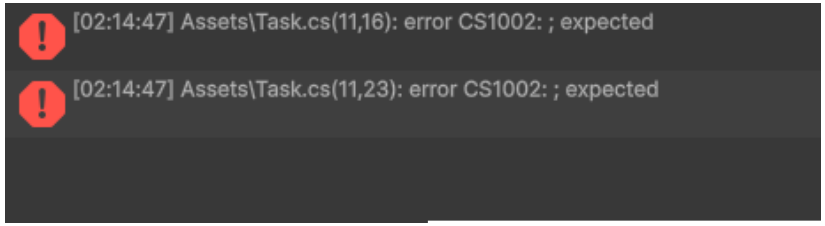



Figure 10: Unity

### 3.4.2 | Godot

Unlike Unity, Godot does have a local editor, which means it can not only tell you what line the bugged code is on, but also highlights it for you (as well as be able to make a better guess as too what's wrong) , as seen in figure 11.

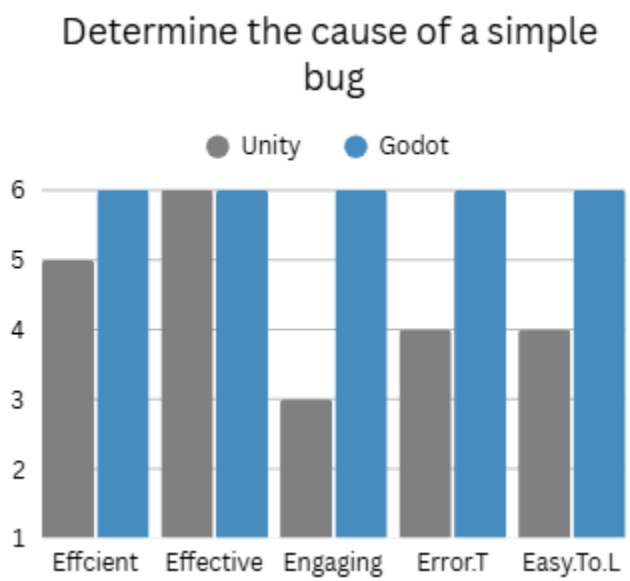



Figure 9: task 4 results bar graph

So, Godot scored much higher overall on this task than Unity due to Godot's built in editor. This is important as developers will spend more time debugging code than actually writing it.

Figure 11: Godot

## 3.5 | Run the scene/project

### 3.5.1 | Unity

Running the scene in Unity was super easy, all I had to do was click the play button on the top middle of the screen then Unity will automatically open the Game window and run the project.

Figure 13: Unity

### 3.5.2 | Godot

Running the scene in Godot is not much different from Unity, however Godot's UI makes it a little difficult to find the correct play button as there are multiple button icons that look like the play button, see figures 14 and 15 to see what I am referencing. For clarity, figure 14 contains Godot's play button.

So, though Godot's UI layout is a bit worse than Unity's in this respect, all it takes it clicking the play button and the project starts playing the scene. As being able to test your project regularly is very important for developers, it is very good that both engines make running your project simple and intuitive.

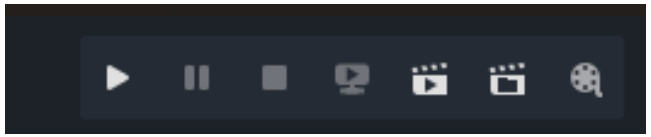


Figure 14: Godot

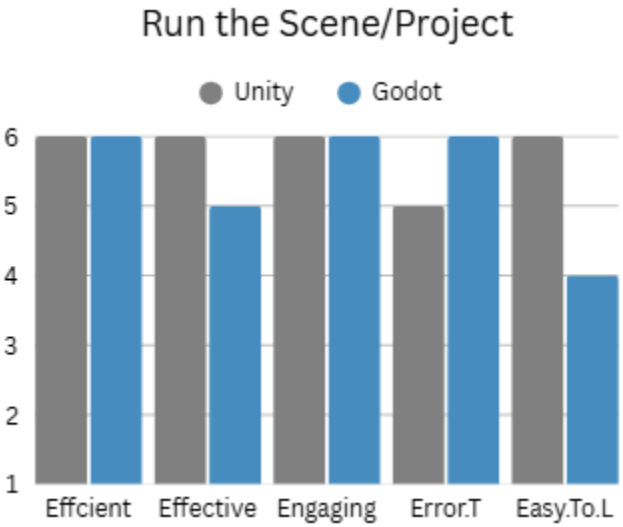



Figure 12: Task 5 results bar chart

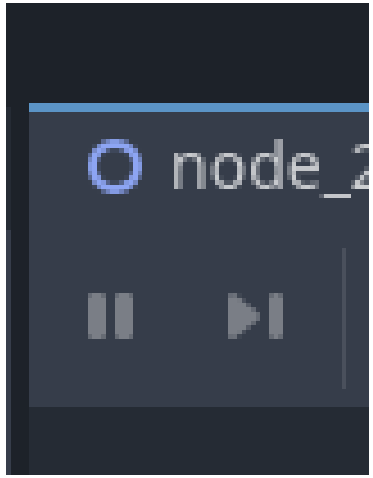



Figure 15: Godot

## 3.6 | Import an Asset

### 3.6.1 | Unity

Unity makes importing assets very easy, it's quite literally just drag and drop. After dragging in the asset Unity provided many options for customizing the asset for my project, see figure 17 for this.

### 3.6.2 | Godot

Godot made importing the asset very difficult. Godot makes all imported asset read-only by default with little to no instruction on how to allow for editing and therefore using an asset. The only help Godot provides is an error message that does not actually offer Specific instruction on how to solve this issue, see figure 18.

So, for this task Unity scored perfectly and Godot scored very low.

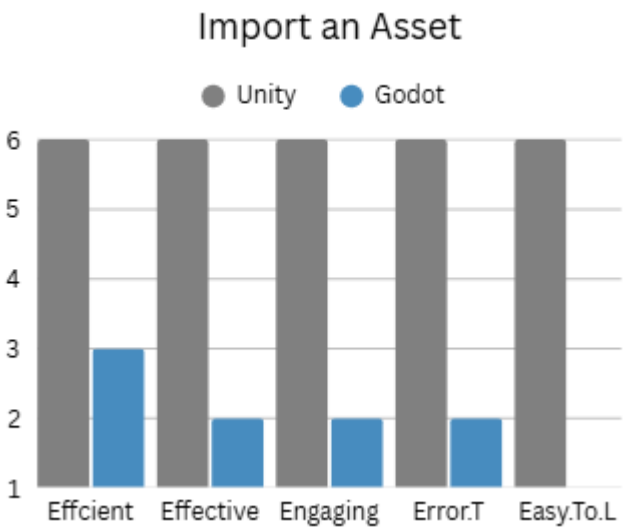



Figure 16: Task 6 results bar chart

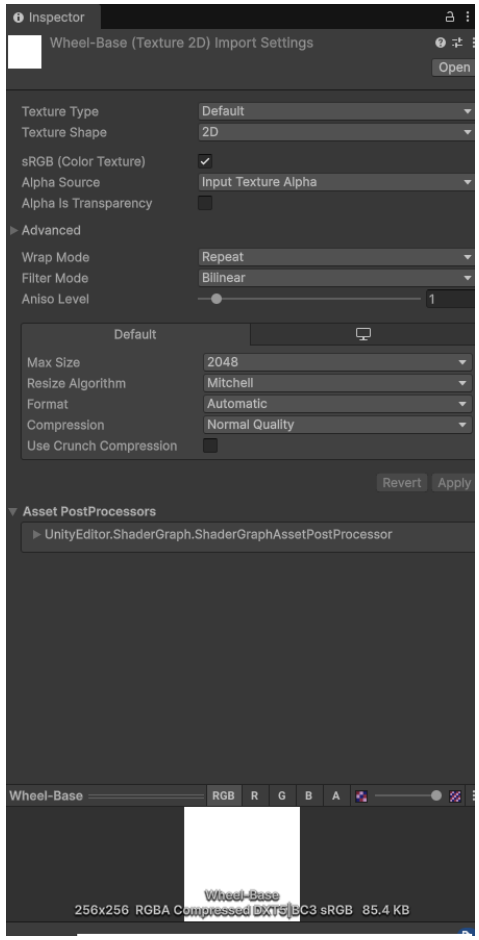



Figure 17: Unity

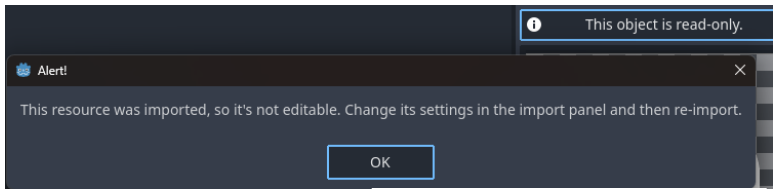



Figure 18: Godot

## 3.7 | Change the Layout

### 3.7.1 | Unity

Overall both engine's make changing the layout of your screen super easy. For Unity all it took was dragging window's to the place I wanted on the screen.

### 3.7.2 | Godot

Godot doesn't support drag and drop like Unity does; but instead uses a drop down menu, as seen in figure 20. For changing the layout I just clicked the vertical ... to get the drop down on a panel, then just selected the position for the panel to jump to, and Godot moved the panel.

Both engine's scored lowly on error tolerance as neither engine provides much direction when there are issue's with window's. That said, both engine's make customizing the layout of your windows/panels super easy.
So, this task's total score mostly came down to personal preference, as both engines support full customization of window/panel customization.

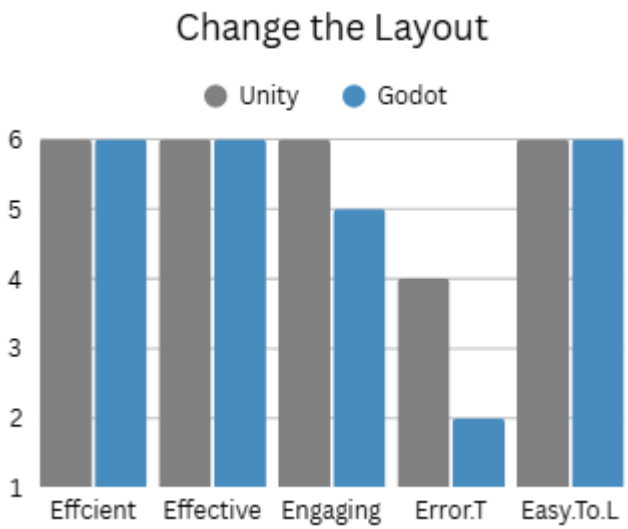



Figure 19: Task 7 results bar chart

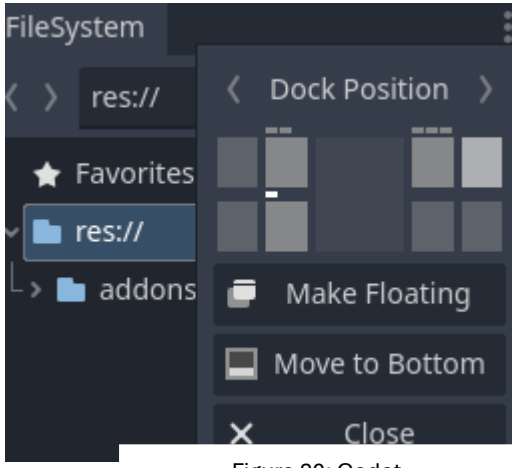



Figure 20: Godot

## 3.8 | Import an Existing Project

### 3.8.1 | Unity

For this task in Unity all it took was clicking Add project from disk in the Add drop down and then locating and opening the project folder, see figure 22. Which is super intuitive and so Unity scored very highly. The low Error tolerant score came from Unity not providing much direction on attempting to open a folder without a valid project, specifically not stating why the project is invalid, see figure 23 for this.

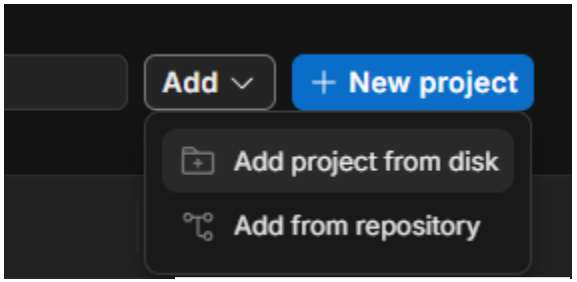



Figure 22: Unity

### 3.8.2 | Godot

Completing this task in Godot was even more streamlined than it was in Unity as Godot has a built-in file explorer which means Godot can give very specific error messages to ensure you open past project's without issue, see figure 24 for an example.

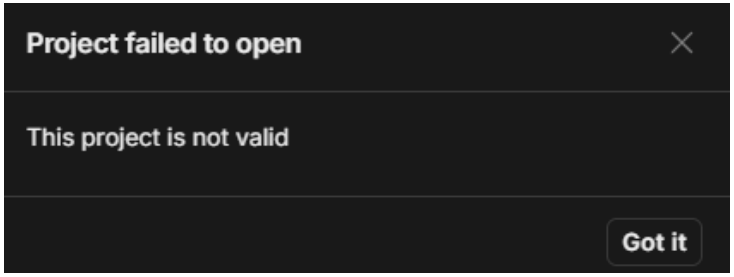



Figure 23: Unity

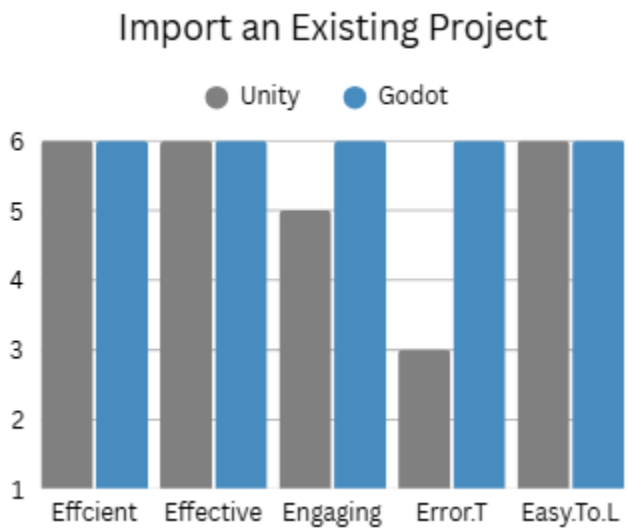



Figure 21: Task 8 results bar chart

So for this task both engine's did well, but Godot made the task even easier with it's built in file explorer.

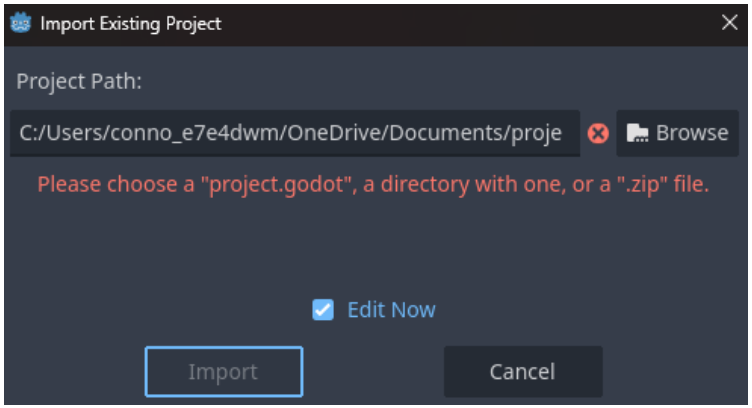



Figure 24: Godot

## 3.9 | Install an Extension

### 3.9.2 | Godot

Completing this task in Godot was actually much easier than it was in Unity. To complete this task all it took was opening Godot's built in Extension search engine, see figure 26. Then just clicking and installing the extension I wanted.

### 3.9.1 | Unity

To complete this task in Unity I could only get extensions are native, that is made by the developer's behind Unity, to Unity within Unity. To do this I opened the package manager selected the package I wanted and installed it, see figure 27.

Godot makes getting and installing extensions very streamlined, simple, and intuitive. Where Unity provides limited options through the package manager.

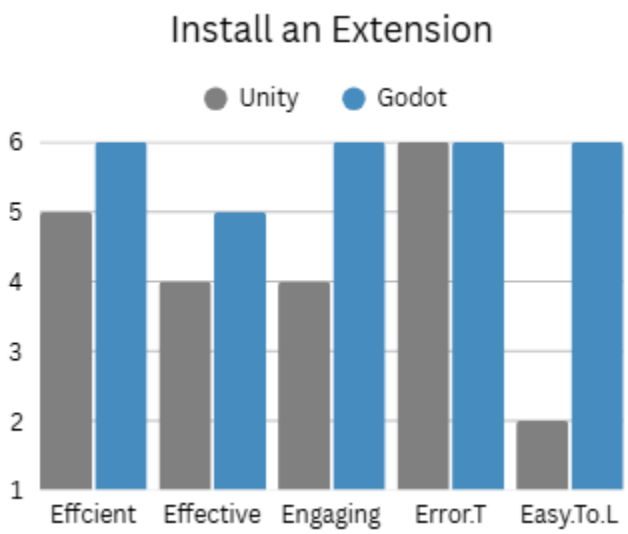



Figure 25: Task 9 results bar chart

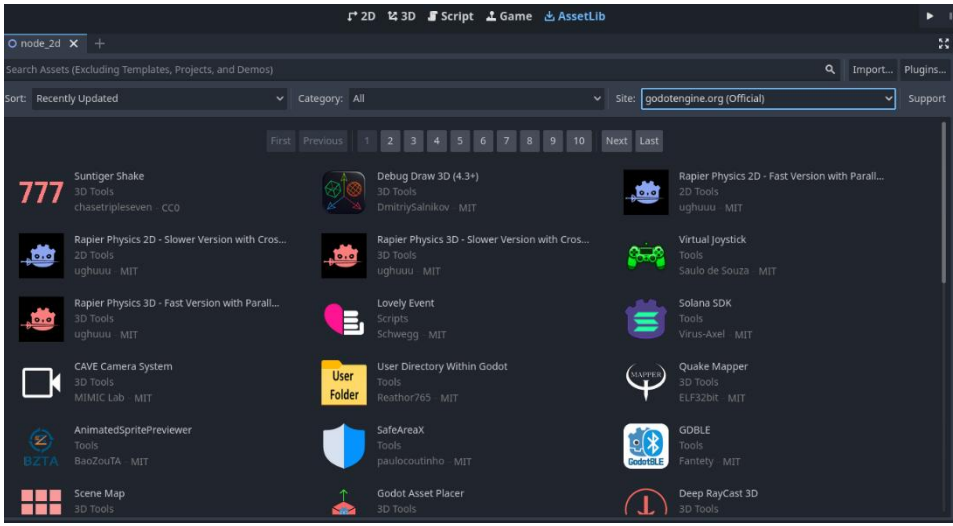



Figure 26: Godot

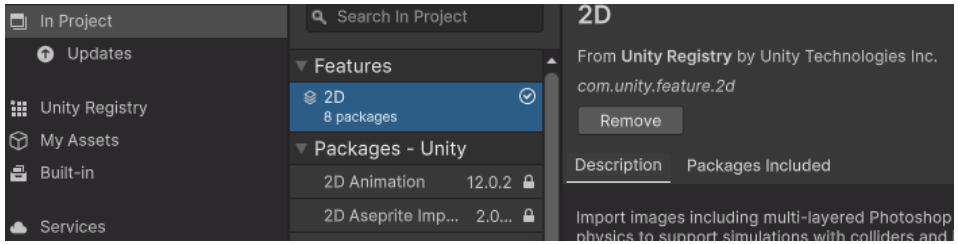



Figure 27: Unity

## 3.10 | Export/Build the Project

### 3.10.1 | Unity

To complete this task in Unity all it took was navigating to build profiles then selecting build. I didn't even need to set the build version as Unity defaults to building for the same platformer as the editor is actively running on, Unity makes building the project very streamlined and simple See figure 29 for the build menu.

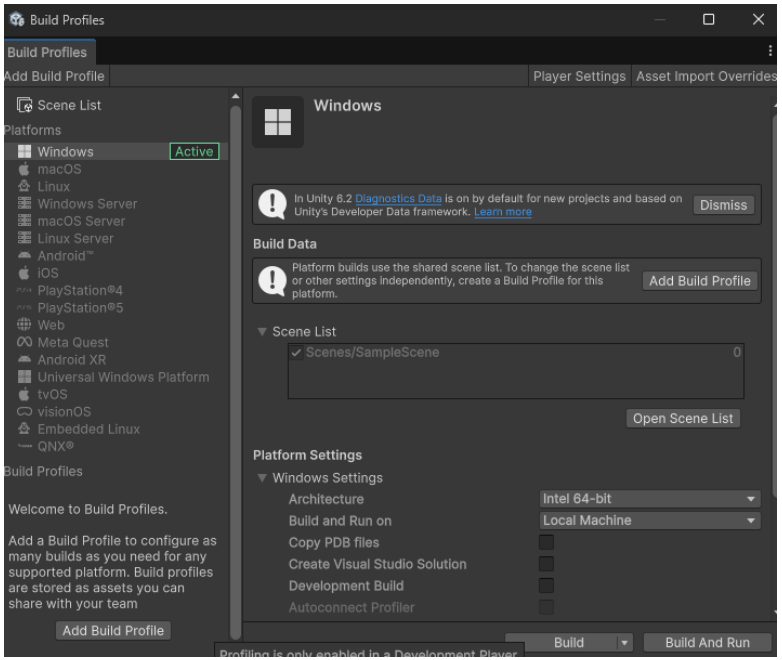



Figure 29: Unity

### 3.10.2 | Godot

To complete this task in Godot it is not much different than in Unity, That said, Godot does awkwardly just close the project after exporting the file without leaving an instruction or notification via a popup or message to ensure me that the task was successful. Additionally, Godot required a little setup to build the project for a specific platform. See figure 30 for Godot's build menu.

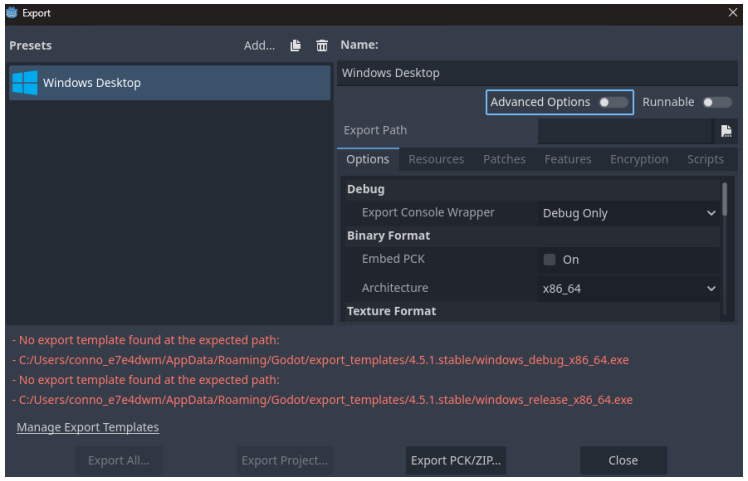



Figure 29: Godot

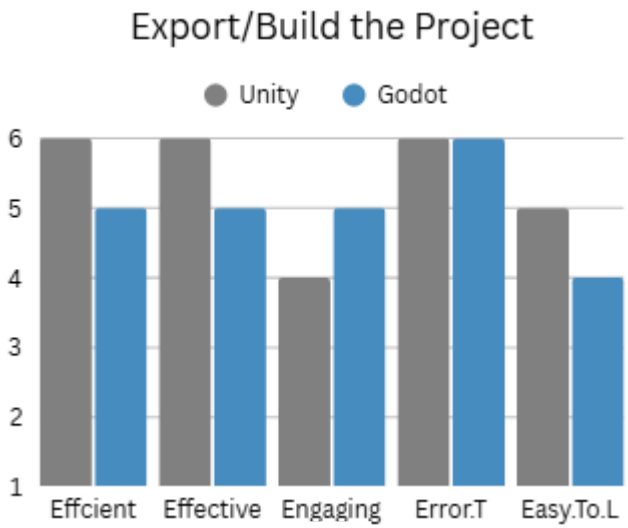



Figure 28: Task 10 results bar chart

Though it's not too difficult to build the project in either engine, it's more streamlined in Unity, and this is why it overall scored higher for this task.

# 4 | Conclusion

**4.1 | Unity Thoughts**

Unity scored higher is most categories due to being more developed and streamlined to be easy to pickup and use. Godot makes starting the project easy with templates and exporting the project easy with its' build tools. And it can't be understated how easy it is to import assets in Unity, especially compared to Godot.

**4.2 | Godot Thoughts**

Though Godot scored lower, it's not by much. And where Godot scored higher individually may influence certain developers to choose Godot over Unity. Specifically Godot provides much more developed support for external extensions that allows developers to add onto Godot and customize the engine to specific development needs.

At the end of the day Unity is licensed software and has been on the market for much longer than Godot. And so, I would recommend Unity over Godot. Unity simply makes the development process easier and most of its downfalls (i.e. debugging) will become less of an issue with general programming and development experience. That said, I still think Godot is fine choice. It's just not as developed as Unity and so has many potentially critical flaws (i.e. importing assets) that may seriously impact development, especially for new developers who already are struggling with learning how to do the already complex task that is game development.

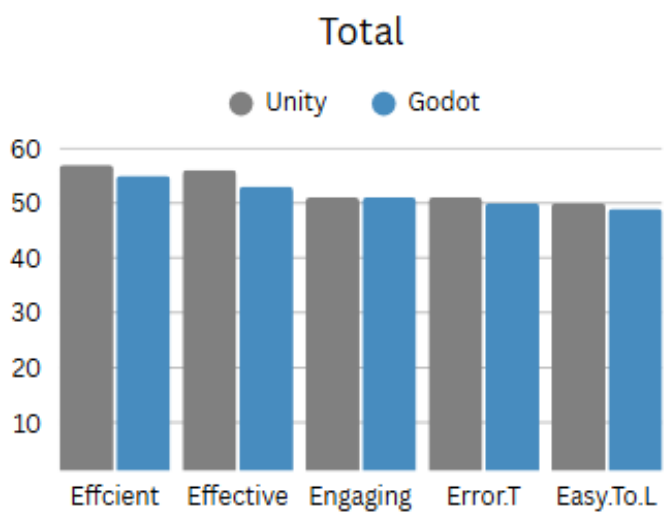



Figure 31: Total Score results bar chart

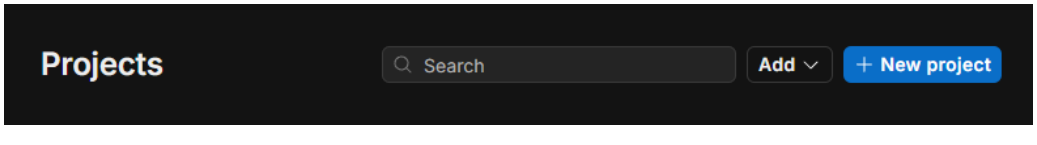
# 5 | Bibliography

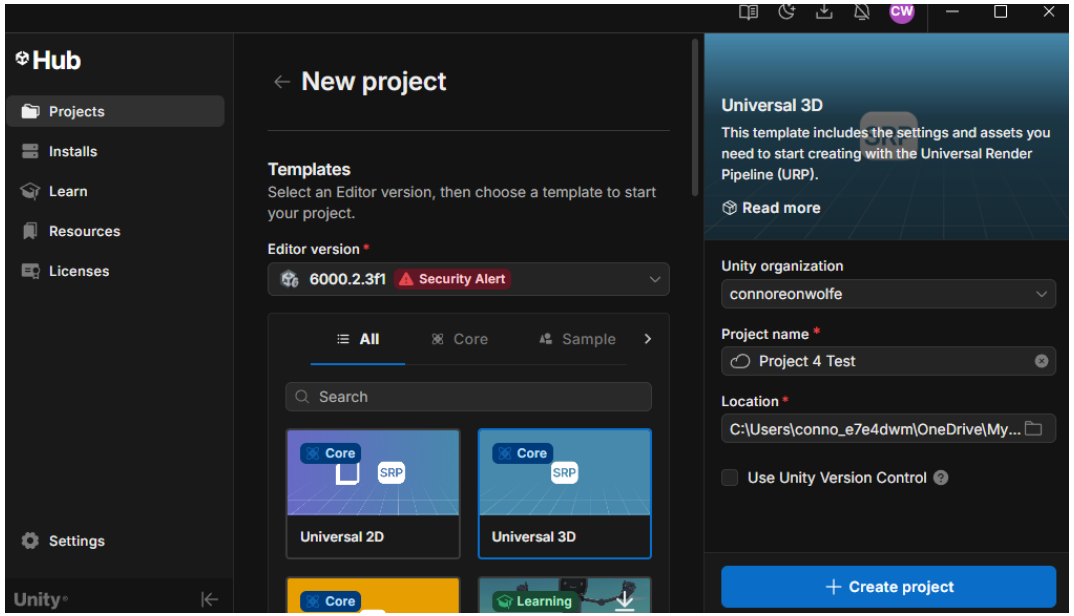
Unity Hub 3.15.1.0
Unity Editor 6000.2.3f1
Godot Editor v4.5.1-stable_win64

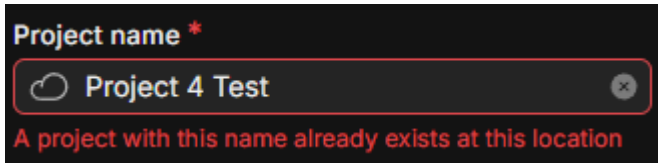
# 6 | Appendix / Documentation

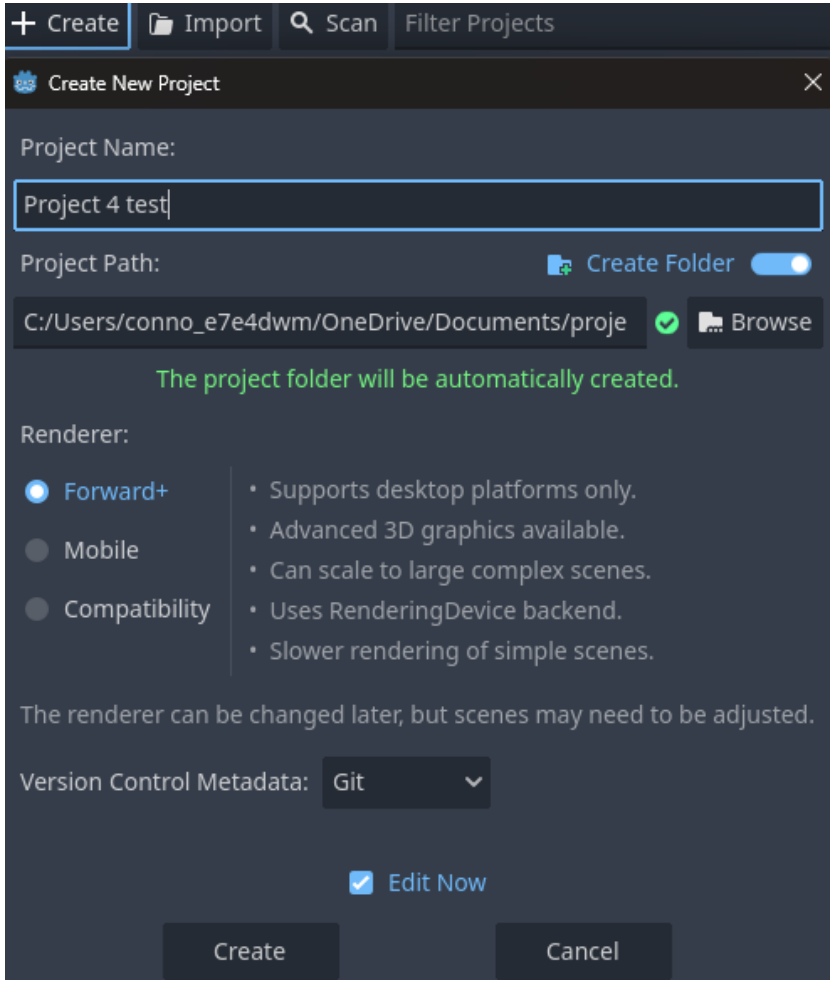
## 6.1 | Create a new project

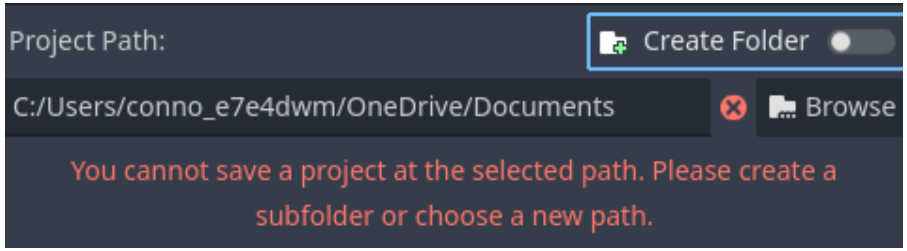
| Tasks | Unity | Godot |
|---|---|---|
| **TASK 1** | Create a new Project | Create a new Project |
| **Efficient** | The task was completed | The task was completed |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Effective** | The task was completed in 3 clicks | The task was completed in 2 clicks |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Engaging** | The Unity UI was very engaging and pleasing | The UI was simple, but a bit plain |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Error Tolerant** | The error message was clear and direct | Gave direct error message |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Easy to Learn** | Templates may be little difficult for the average user, but not too un-intuitive | All buttons were clear and needed no instructions |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Additional Notes** | To create a new project I clicked "+ New Project", then I had to select what template this project will use, after selecting a template (in this case "Universal 3D") I clicked "+ Create project". After some loading, the project opened.<br>Should I name the project with the same name as another project in that location, an error message informs the users of this issue | To complete this task I clicked "Create", then clicked create on the next window.<br>Should I choose an invalid path to save the project, a informational error message pops up to inform the user on the issue |


Projects
Search
Add
New project


Hub
Projects
Installs
Learn
Resources
Licenses
Settings
Unity
New project
Templates
Select an Editor version, then choose a template to start your project.
Editor version *
6000.2.3f1
Security Alert
All
Core
Sample
Search
Core
SRP
Universal 2D
Core
SRP
Universal 3D
Core
Learning
Universal 3D
This template includes the settings and assets you need to start creating with the Universal Render Pipeline (URP).
Read more
Unity organization
connoreonwolfe
Project name *
Project 4 Test
Location *
C:\Users\conno_e7e4dwm\OneDrive\My...
Use Unity Version Control
Create project

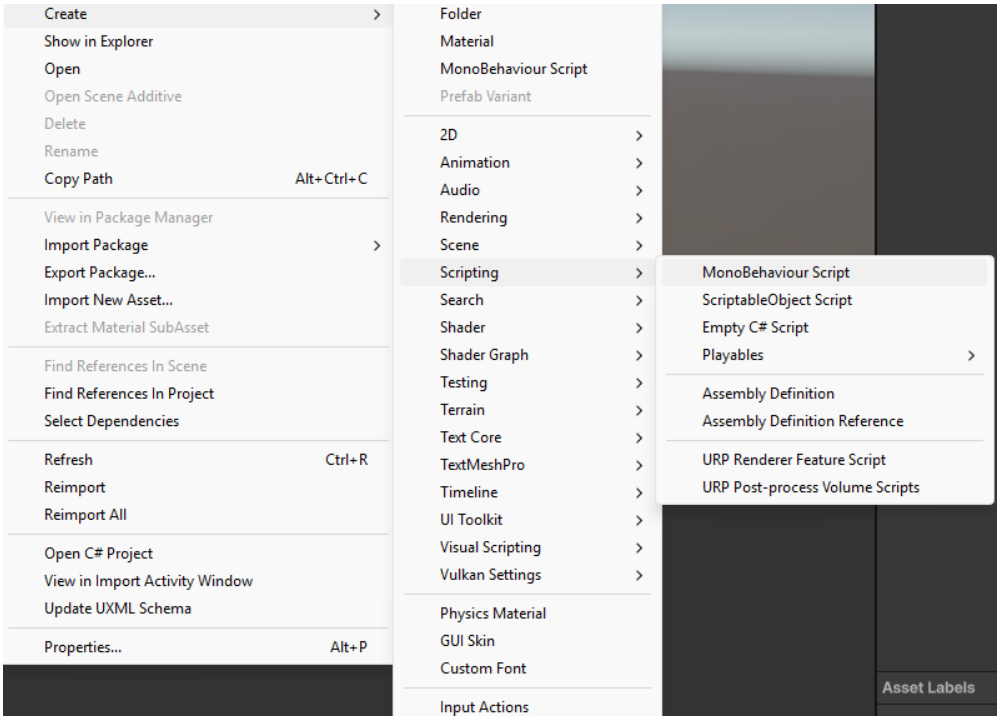

Project name *
Project 4 Test
A project with this name already exists at this location

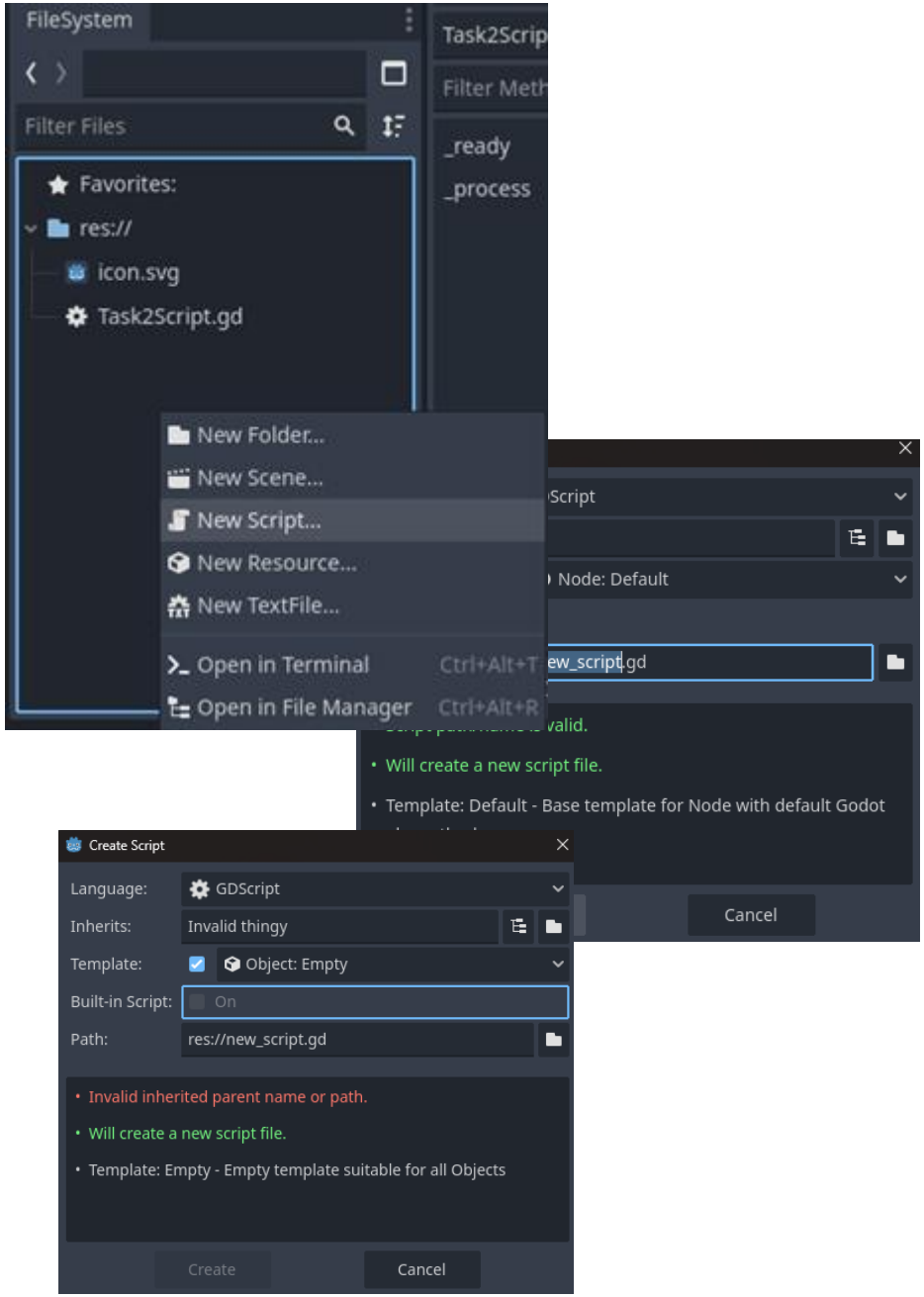

Create
Import
Scan
Filter Projects
Create New Project
Project Name:
Project 4 test
Project Path:
Create Folder
C:/Users/conno_e7e4dwm/OneDrive/Documents/proje
Browse
The project folder will be automatically created.
Renderer:
Forward+
Mobile
Compatibility
Supports desktop platforms only.
Advanced 3D graphics available.
Can scale to large complex scenes.
Uses RenderingDevice backend.
Slower rendering of simple scenes.
The renderer can be changed later, but scenes may need to be adjusted.
Version Control Metadata:
Git
Edit Now
Create
Cancel


Project Path:
Create Folder
C:/Users/conno_e7e4dwm/OneDrive/Documents
Browse
You cannot save a project at the selected path. Please create a subfolder or choose a new path.

## 6.2 | Create a script

| Tasks | Unity | Godot |
|---|---|---|
| **TASK 2** | Create a script | Create a script |
| **Efficient** | I was able to successfully create a gd script | I was able to successfully create a gd script |
| | **Strongly Disagree** / **Disagree** / **Slightly Disagree** / **Slightly Agree** / **Agree** / **Strongly Agree**<br>**1** / 2 / 3 / 4 / 5 / 6 | **Strongly Disagree** / **Disagree** / **Slightly Disagree** / **Slightly Agree** / **Agree** / **Strongly Agree**<br>**1** / 2 / 3 / 4 / 5 / 6 |
| **Effective** | It took 4 clicks (3 if you take the presented shortcut) | It took 2 clicks |
| | **Strongly Disagree** / **Disagree** / **Slightly Disagree** / **Slightly Agree** / **Agree** / **Strongly Agree**<br>**1** / 2 / 3 / 4 / 5 / 6 | **Strongly Disagree** / **Disagree** / **Slightly Disagree** / **Slightly Agree** / **Agree** / **Strongly Agree**<br>**1** / 2 / 3 / 4 / 5 / 6 |
| **Engaging** | The UI was straight forward and clean | The UI was straight forward and clean |
| | **Strongly Disagree** / **Disagree** / **Slightly Disagree** / **Slightly Agree** / **Agree** / **Strongly Agree**<br>**1** / 2 / 3 / 4 / 5 / 6 | **Strongly Disagree** / **Disagree** / **Slightly Disagree** / **Slightly Agree** / **Agree** / **Strongly Agree**<br>**1** / 2 / 3 / 4 / 5 / 6 |
| **Error Tolerant** | Unity took care of common errors cleanly in a way that was unintrusive | The error message was a little ambiguous |
| | **Strongly Disagree** / **Disagree** / **Slightly Disagree** / **Slightly Agree** / **Agree** / **Strongly Agree**<br>**1** / 2 / 3 / 4 / 5 / 6 | **Strongly Disagree** / **Disagree** / **Slightly Disagree** / **Slightly Agree** / **Agree** / **Strongly Agree**<br>**1** / 2 / 3 / 4 / 5 / 6 |
| **Easy to Learn** | There were a lot of steps to create a script, which may have taken a little bit of navigation | The actions to create a script are the standard expected actions in development software |
| | **Strongly Disagree** / **Disagree** / **Slightly Disagree** / **Slightly Agree** / **Agree** / **Strongly Agree**<br>**1** / 2 / 3 / 4 / 5 / 6 | **Strongly Disagree** / **Disagree** / **Slightly Disagree** / **Slightly Agree** / **Agree** / **Strongly Agree**<br>**1** / 2 / 3 / 4 / 5 / 6 |
| **Additional Notes** | To create a script I had to right click in the project window and then click “Create”->”Scripting”->”MonoBehavior Script”.<br>I you make multiple scripts with the same name, Unity will simply concatenate on a number, and increment it. | To create a script all I had to do was a right click and then click “New Script…” (optionally name the script, manually select a file location, and/or set certain file settings), then click “Create”.<br>Should the files path or inheritance be incorrectly set it displays the error “Invalid inherited parent name or path” which is a little ambiguous. A unique error message for either case would have been more optimal. |

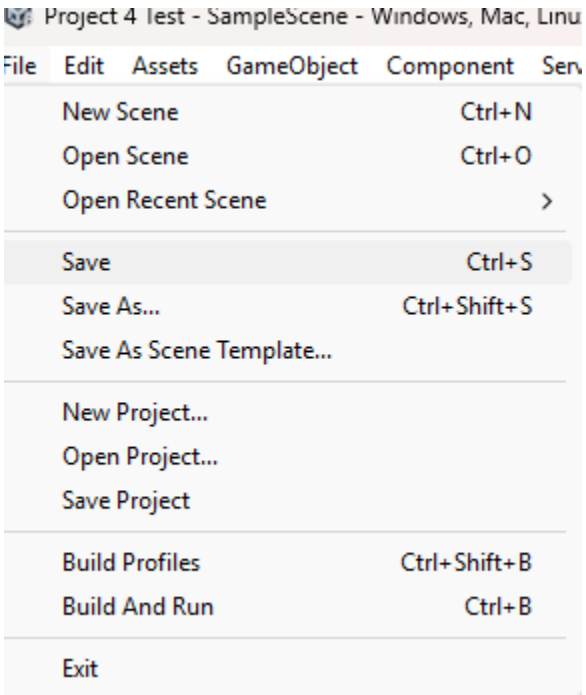

Create
Show in Explorer
Open
Open Scene Additive
Delete
Rename
Copy Path Alt+Ctrl+C
View in Package Manager
Import Package
Export Package...
Import New Asset...
Extract Material SubAsset
Find References In Scene
Find References In Project
Select Dependencies
Refresh Ctrl+R
Reimport
Reimport All
Open C# Project
View in Import Activity Window
Update UXML Schema
Properties... Alt+P
Folder
Material
MonoBehaviour Script
Prefab Variant
2D
Animation
Audio
Rendering
Scene
Scripting
Search
Shader
Shader Graph
Testing
Terrain
Text Core
TextMeshPro
Timeline
UI Toolkit
Visual Scripting
Vulkan Settings
Physics Material
GUI Skin
Custom Font
Input Actions
MonoBehaviour Script
ScriptableObject Script
Empty C# Script
Playables
Assembly Definition
Assembly Definition Reference
URP Renderer Feature Script
URP Post-process Volume Scripts
Asset Labels

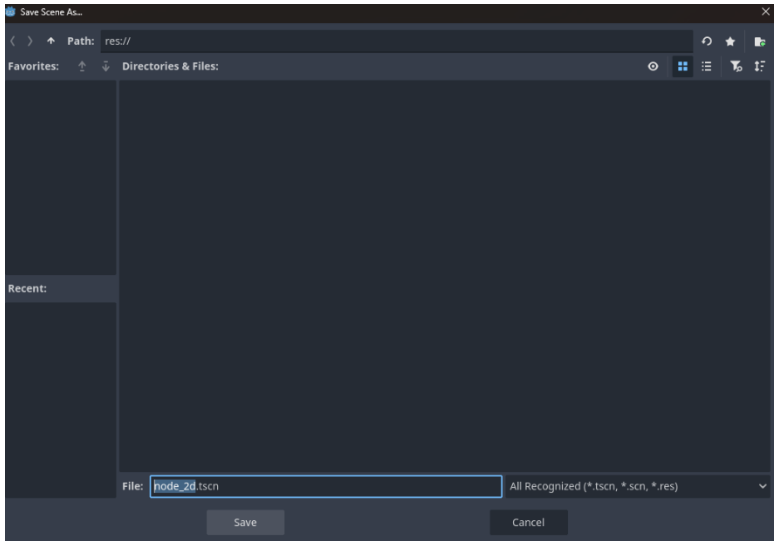

FileSystem
Filter Files
Favorites:
res://
icon.svg
Task2Script.gd
New Folder...
New Scene...
New Script...
New Resource...
New TextFile...
Open in Terminal Ctrl+Alt+T
Open in File Manager Ctrl+Alt+R
Task2Scrip
Filter Meth
_ready
_process
Script
Node: Default
ew_script.gd
Will create a new script file.
Template: Default - Base template for Node with default Godot
Cancel
Create Script
Language: GDScript
Inherits: Invalid thingy
Template: Object: Empty
Built-in Script: On
Path: res://new_script.gd
Invalid inherited parent name or path.
Will create a new script file.
Template: Empty - Empty template suitable for all Objects
Create
Cancel

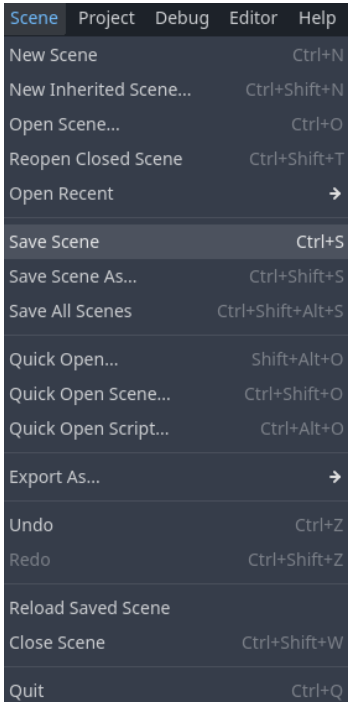
Task
Task1
Task2
Task3

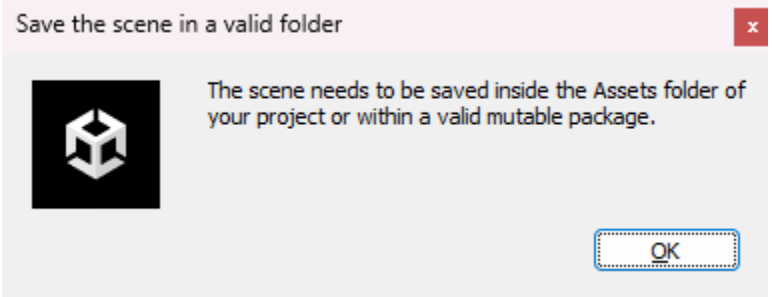
## 6.3 | Save the Project Files

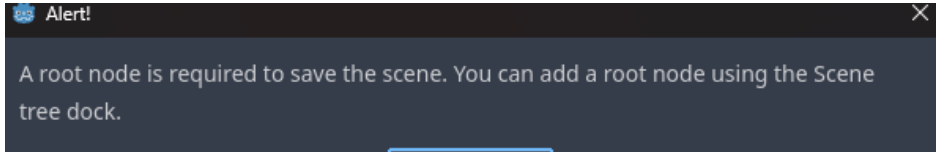
| Tasks | Unity | Godot |
|---|---|---|
| **TASK 3** | Save Project Files | Save Project Files |
| **Efficient** | I was able to complete the task | I was able to complete the task with some difficulty |
| | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Effective** | It took 2 clicks | It took 3 clicks |
| | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Engaging** | The UI effectively communicates how/where to save | The UI does not communicate how to do this action well |
| | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Error Tolerant** | The error message was clear and gave direct instructions | The error message was clear and gave direct instructions |
| | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Easy to Learn** | Saving the project works just as it should, very intuitive | Scene == Project when there is also a project tab is not intuitive |
| | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Additional Notes** | To save the project I clicked “File”->”Save”<br><br>Should I click “File”->”Save As…” and try to save the file outside the project, I get a popup to inform the user that the save needs to be done within the file | To save the project (the first time) you have to click Scene -> ”Save Scene” then select a file name and location then click “Save” . This was confusing as there is a Project tab as well. The terminology ‘Scene’ for these kinds of programs does not typically equate to project, so this was confusing and not intuitive at all.<br>If you do not have a root node, a very good error message appears to inform you. |


Project 4 Test - SampleScene - Windows, Mac, Linu
File Edit Assets GameObject Component Serv
New Scene Ctrl+N
Open Scene Ctrl+O
Open Recent Scene
Save Ctrl+S
Save As... Ctrl+Shift+S
Save As Scene Template...
New Project...
Open Project...
Save Project
Build Profiles Ctrl+Shift+B
Build And Run Ctrl+B
Exit


Save Scene As...
Path: res://
Favorites:
Directories & Files:
Recent:
File: node_2d.tscn
All Recognized (*.tscn, *.scn, *.res)
Save
Cancel

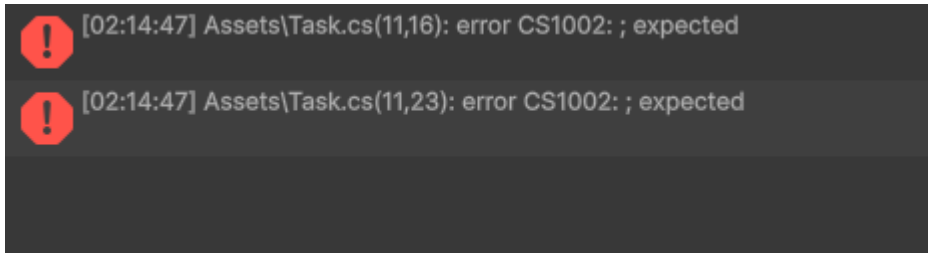

Scene Project Debug Editor Help
New Scene Ctrl+N
New Inherited Scene... Ctrl+Shift+N
Open Scene... Ctrl+O
Reopen Closed Scene Ctrl+Shift+T
Open Recent
Save Scene Ctrl+S
Save Scene As... Ctrl+Shift+S
Save All Scenes Ctrl+Shift+Alt+S
Quick Open... Shift+Alt+O
Quick Open Scene... Ctrl+Shift+O
Quick Open Script... Ctrl+Alt+O
Export As...
Undo Ctrl+Z
Redo Ctrl+Shift+Z
Reload Saved Scene
Close Scene Ctrl+Shift+W
Quit Ctrl+Q

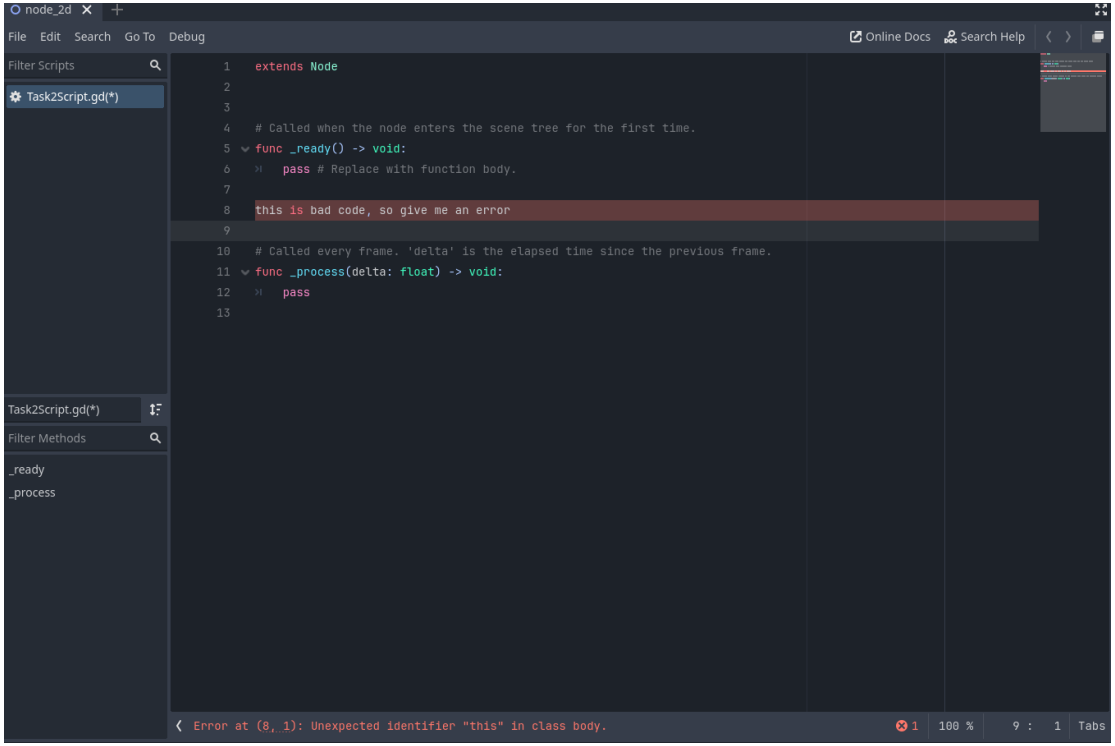

Save the scene in a valid folder
The scene needs to be saved inside the Assets folder of your project or within a valid mutable package.
OK


Alert!
A root node is required to save the scene. You can add a root node using the Scene tree dock.

## 6.4 | Determine the cause of a simple bug

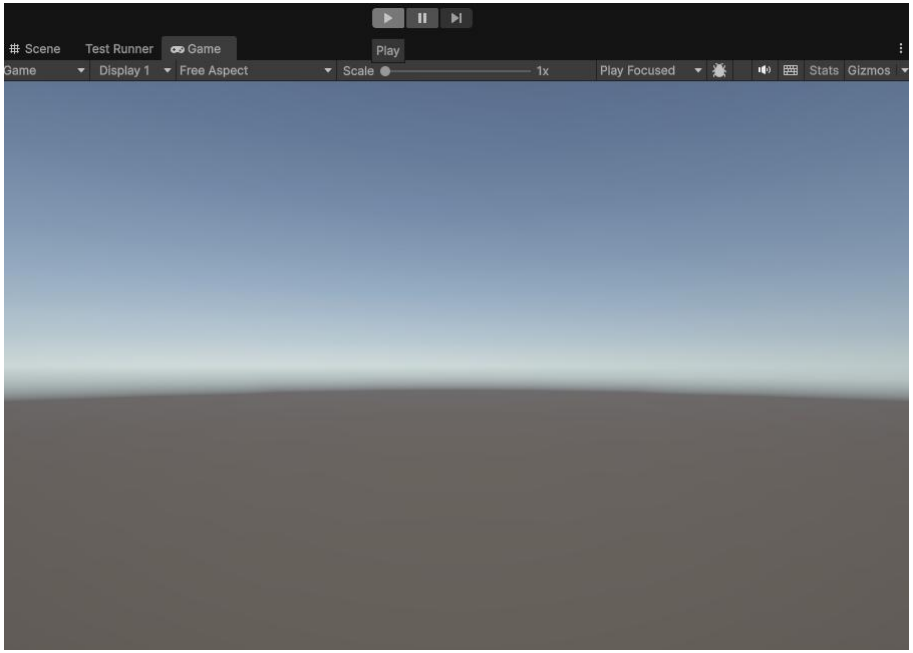
| Tasks | Unity | Godot |
|---|---|---|
| **TASK 4** | Determine the cause of a simple bug | Determine the cause of a simple bug |
| **Efficient** | This task was completed, but took some navigation | This task was completed |
| | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Effective** | The error appeared as soon as I wrote the bad code | The error appeared as soon as I wrote the bad code |
| | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Engaging** | The console window can be a little difficult to use | Godot highlighted the line of bugged code |
| | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Error Tolerant** | Unity provided error messages in the console window communicating where the error <u>might</u> be, to help your search | Exact line numbers and a description of the potential issue was given |
| | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Easy to Learn** | Unity provides line number to help, but is not specific on what the error is, which means it takes some problem solving on the developer's end | The error detection makes it very clear where the bugged code is, and what the issue might be |
| | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / **Disagree** 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Additional Notes** | To test this task I wrote a bogus line of code in a script to see how unity handles this. Unity displays error message in the console window describing the potential issue, and what line(s) it could have occurred on. But as Unity does have a native code editor, it cannot provide the specifics on what exactly the code bug is. | To test this task I wrote a bogus line of code to see how Godot handle's code bugs. It highlighted the line and gave an error message with line numbers as soon as I finished the line. |

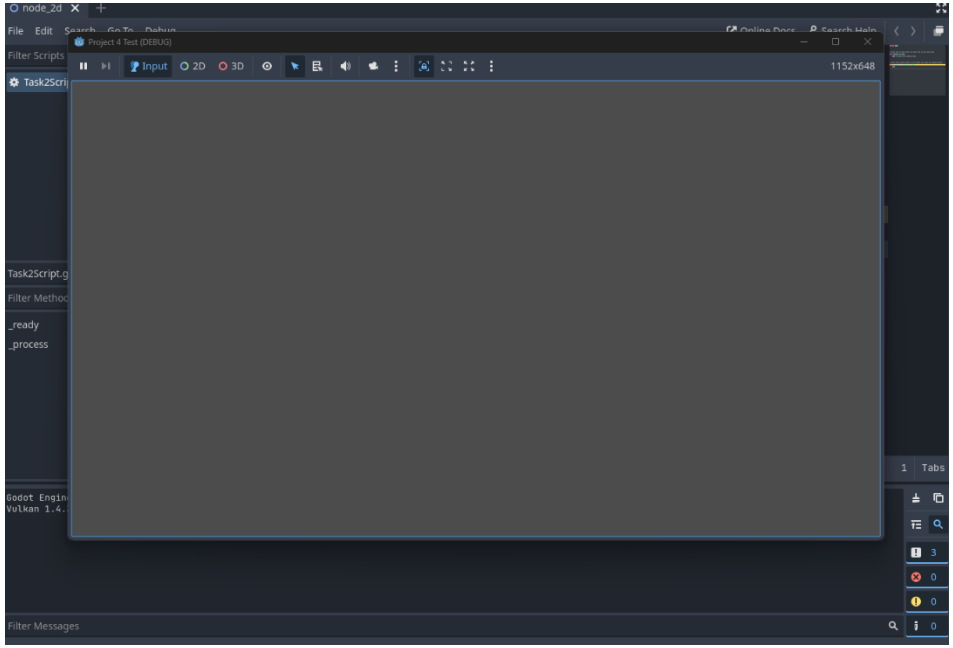

[02:14:47] Assets\Task.cs(11,16): error CS1002: ; expected
[02:14:47] Assets\Task.cs(11,23): error CS1002: ; expected

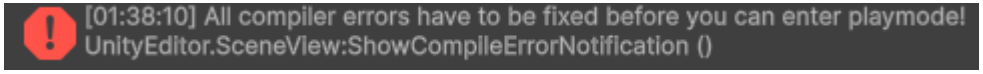

node_2d
File Edit Search Go To Debug
Online Docs
Search Help
Filter Scripts
Task2Script.gd(*)
extends Node
# Called when the node enters the scene tree for the first time.
func _ready() -> void:
pass # Replace with function body.
this is bad code, so give me an error
# Called every frame. 'delta' is the elapsed time since the previous frame.
func _process(delta: float) -> void:
pass
Task2Script.gd(*)
Filter Methods
_ready
_process
Error at (8, 1): Unexpected identifier "this" in class body.
1
100 %
9 : 1
Tabs

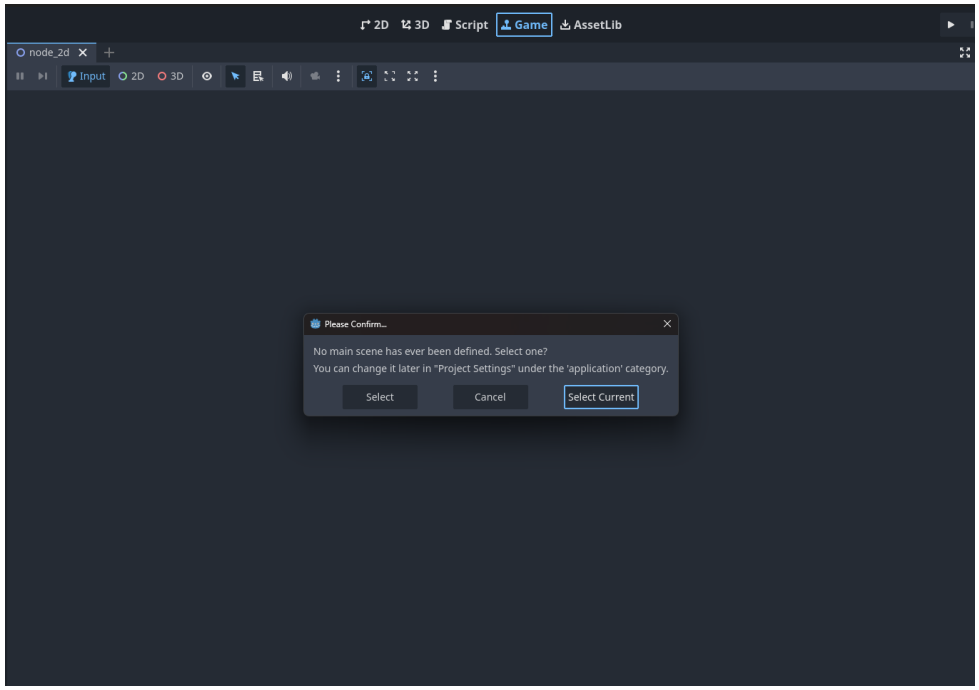
## 6.5 | Run the scene/project

| Tasks | Unity | Godot |
|---|---|---|
| **TASK 5** | Run the Scene/Project | Run the Scene/Project |
| **Efficient** | The task was completed | The task was completed |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Effective** | The task was completed in one click | The task was completed in two clicks |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Engaging** | The UI is affective | The UI is simple but effective |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Error Tolerant** | The error message are a little out of the way (first on the most bottom sliver of the screen, then also in the console window if you know to look there), but give direct and helpful information | Gives effective prompts in cases of missing selections |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Easy to Learn** | The task was very intuitive, just click the play button in the top middle of the screen | The play button was a little out of the way, and there were some other buttons that looked like they might be play button |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Additional Notes** | To run the scene I clicked the play button at the top middle section of the screen, after some loading the scene ran. If the Game window isn't open when you click Play, it automatically opens to show you the game scene.<br>Should there be compiler errors and the scene can't be run, there are helpful error message in the console window | To run the scene I went to the "Game Tab", clicked the play button on the top left. Then, the scene ran<br>Since there was no main scene selected it prompted me to select a scene to become the main scene. |


Scene
Test Runner
Game
Play
Game
Display 1
Free Aspect
Scale
1x
Play Focused
Stats
Gizmos

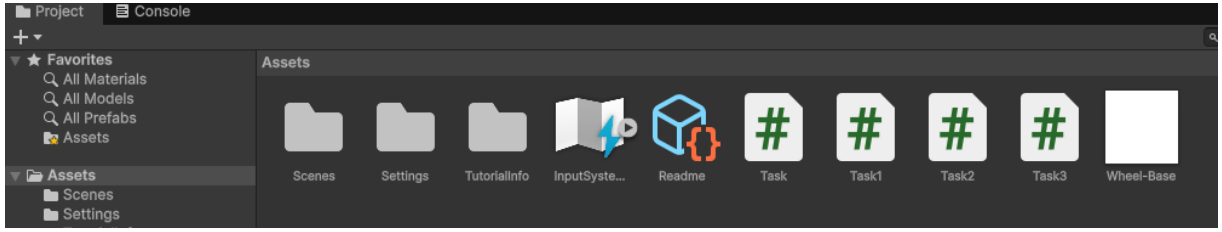

node_2d
File
Edit
Project 4 Test (DEBUG)
Input
2D
3D
1152x648
Filter Scripts
Task2Scri
Task2Script.g
Filter Methoc
_ready
_process
Godot Engin
Vulkan 1.4.
1
Tabs
Filter Messages

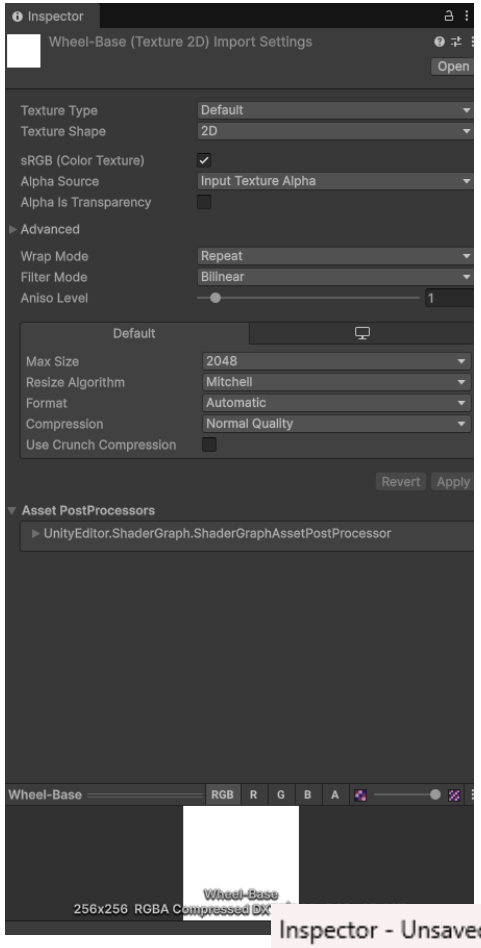

[01:38:10] All compiler errors have to be fixed before you can enter playmode!
UnityEditor.SceneView:ShowCompileErrorNotification ()

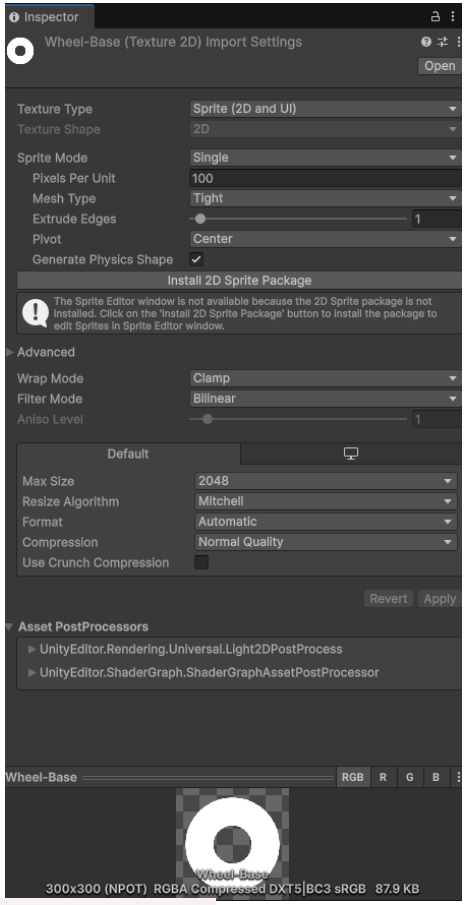

2D
3D
Script
Game
AssetLib
node_2d
Input
2D
3D
Please Confirm...
No main scene has ever been defined. Select one?
You can change it later in "Project Settings" under the 'application' category.
Select
Cancel
Select Current

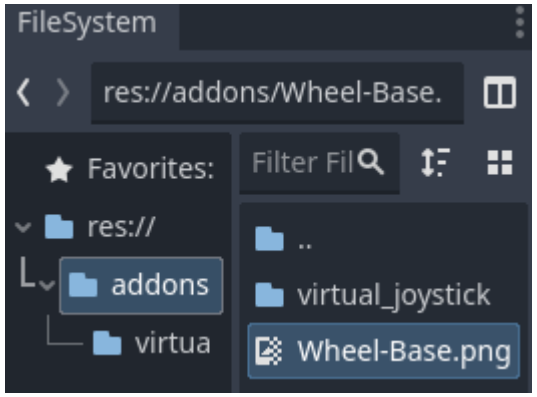
## 6.6 | Import an Asset

| Tasks | Unity | Godot |
|---|---|---|
| **TASK 6** | Import an asset | Import an asset |
| **Efficient** | I was able to add a png asset to the game | I was able to add a png to the game, but was unable to make it writable/changable |
| | **Strongly Disagree** **1**, 2, **Disagree**, 3, **Slightly Disagree**, 4, **Slightly Agree**, 5, **Agree**, 6, **Strongly Agree** | **Strongly Disagree** **1**, 2, **Disagree**, 3, **Slightly Disagree**, 4, **Slightly Agree**, 5, **Agree**, 6, **Strongly Agree** |
| **Effective** | Importing an asset requires drag and drop (with minor tuning in the inspector depending on the asset) | Importing an asset requires more steps than just drag an drop. |
| | **Strongly Disagree** **1**, 2, **Disagree**, 3, **Slightly Disagree**, 4, **Slightly Agree**, 5, **Agree**, 6, **Strongly Agree** | **Strongly Disagree** **1**, 2, **Disagree**, 3, **Slightly Disagree**, 4, **Slightly Agree**, 5, **Agree**, 6, **Strongly Agree** |
| **Engaging** | Adding an asset was nearly seamless | Importing the asset was not engaging |
| | **Strongly Disagree** **1**, 2, **Disagree**, 3, **Slightly Disagree**, 4, **Slightly Agree**, 5, **Agree**, 6, **Strongly Agree** | **Strongly Disagree** **1**, 2, **Disagree**, 3, **Slightly Disagree**, 4, **Slightly Agree**, 5, **Agree**, 6, **Strongly Agree** |
| **Error Tolerant** | I helpful and hard to ignore popup occurs when leaving the inspector | Godot gave little instruction on how to change the asset from read-only |
| | **Strongly Disagree** **1**, 2, **Disagree**, 3, **Slightly Disagree**, 4, **Slightly Agree**, 5, **Agree**, 6, **Strongly Agree** | **Strongly Disagree** **1**, 2, **Disagree**, 3, **Slightly Disagree**, 4, **Slightly Agree**, 5, **Agree**, 6, **Strongly Agree** |
| **Easy to Learn** | Drag and Drop | Importing an asset should not take many more steps than drag and drop, before you can edit the asset. |
| | **Strongly Disagree** **1**, 2, **Disagree**, 3, **Slightly Disagree**, 4, **Slightly Agree**, 5, **Agree**, 6, **Strongly Agree** | **Strongly Disagree** **1**, 2, **Disagree**, 3, **Slightly Disagree**, 4, **Slightly Agree**, 5, **Agree**, 6, **Strongly Agree** |
| **Additional Notes** | To import an asset, I dragged a png asset into the project window. Then I was free to select any desired import settings to change the asset how I'd like.<br>Should you leave the inspector with unsaved changes a helpful popup occurs informing you and giving you options on what to do next | To complete this task I dragged into the "Filesystem" window a simple png asset. Then I had to make the png file writable as imported assets are read-only by default.<br>After testing every setting in the Import window I could not figure out how to make the object writable.<br>So it took googling to figure out why it was this way, and it turns out: Godot makes imported assets read only. To get a writable version, I would have to make another copy of the asset within Godot for Godot to let me write to the asset. |

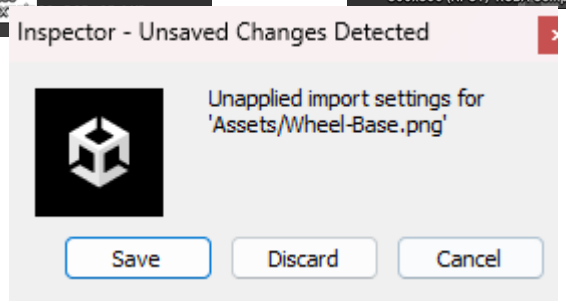

Project
Console
Favorites
All Materials
All Models
All Prefabs
Assets
Assets
Scenes
Settings
Scenes
Settings
TutorialInfo
InputSyste...
Readme
Task
Task1
Task2
Task3
Wheel-Base

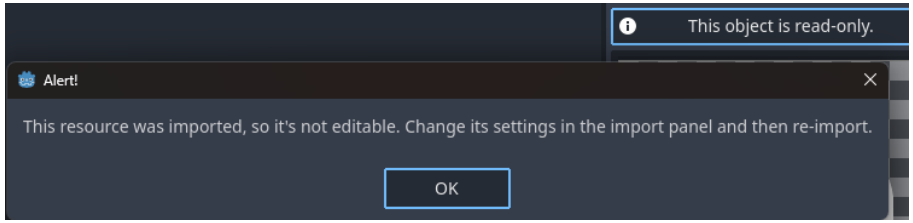

Inspector
Wheel-Base (Texture 2D) Import Settings
Open
Texture Type
Default
Texture Shape
2D
sRGB (Color Texture)
Alpha Source
Input Texture Alpha
Alpha Is Transparency
Advanced
Wrap Mode
Repeat
Filter Mode
Bilinear
Aniso Level
1
Default
Max Size
2048
Resize Algorithm
Mitchell
Format
Automatic
Compression
Normal Quality
Use Crunch Compression
Revert
Apply
Asset PostProcessors
UnityEditor.ShaderGraph.ShaderGraphAssetPostProcessor
Wheel-Base
RGB
R
G
B
A
256x256 RGBA Compressed DX


Inspector
Wheel-Base (Texture 2D) Import Settings
Open
Texture Type
Sprite (2D and UI)
Texture Shape
2D
Sprite Mode
Single
Pixels Per Unit
100
Mesh Type
Tight
Extrude Edges
1
Pivot
Center
Generate Physics Shape
Install 2D Sprite Package
The Sprite Editor window is not available because the 2D Sprite package is not installed. Click on the 'Install 2D Sprite Package' button to install the package to edit Sprites in Sprite Editor window.
Advanced
Wrap Mode
Clamp
Filter Mode
Bilinear
Aniso Level
1
Default
Max Size
2048
Resize Algorithm
Mitchell
Format
Automatic
Compression
Normal Quality
Use Crunch Compression
Revert
Apply
Asset PostProcessors
UnityEditor.Rendering.Universal.Light2DPostProcess
UnityEditor.ShaderGraph.ShaderGraphAssetPostProcessor
Wheel-Base
RGB
R
G
B
300x300 (NPOT) RGBA Compressed DXT5|BC3 sRGB 87.9 KB


FileSystem
res://addons/Wheel-Base.
Favorites:
Filter Fil
res://
addons
virtua
..
virtual_joystick
Wheel-Base.png

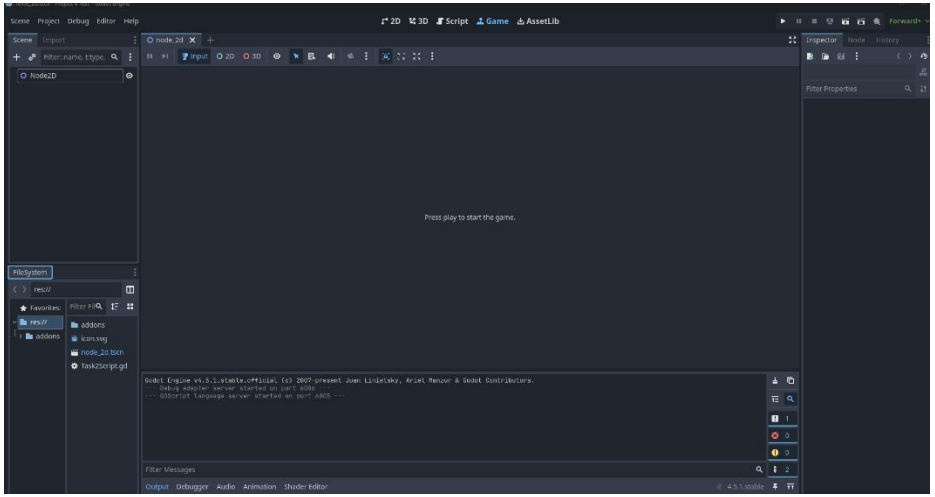
Inspector
Node
History
Wheel-Base.png
Filter Properties
This object is read-only.
300×300 RGBA8
No Mipmaps
Memory: 351.5 KiB
CompressedTexture2D
Load Path
res://.godot
Resource
Resource
RefCounted
Script
<empty>
Add Metadata

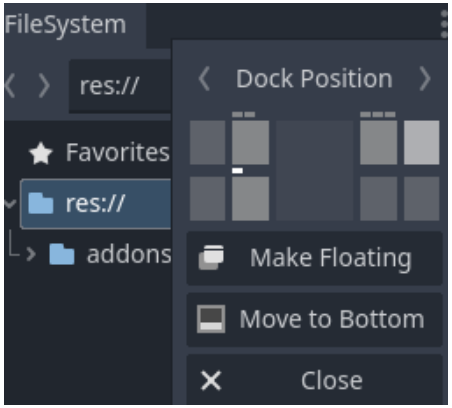

Inspector - Unsaved Changes Detected
Unapplied import settings for
'Assets/Wheel-Base.png'
Save
Discard
Cancel

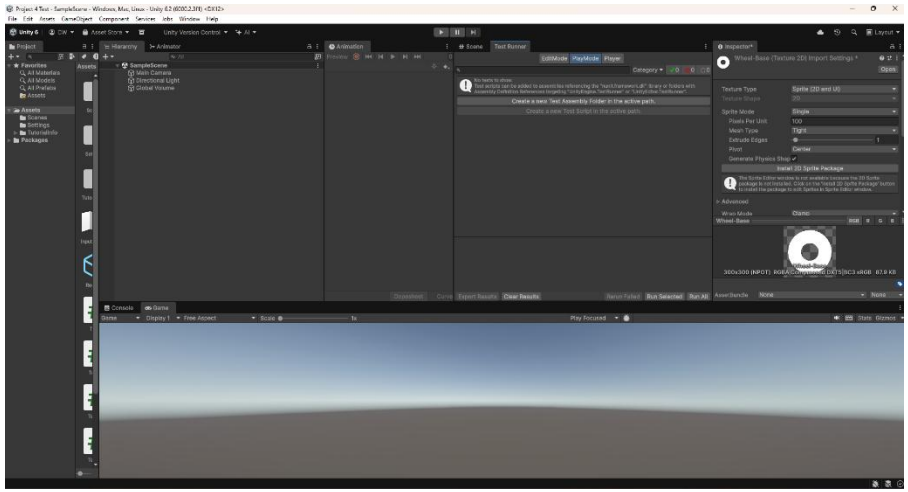

This object is read-only.
Alert!
This resource was imported, so it's not editable. Change its settings in the import panel and then re-import.
OK

## 6.7 | Change the Layout

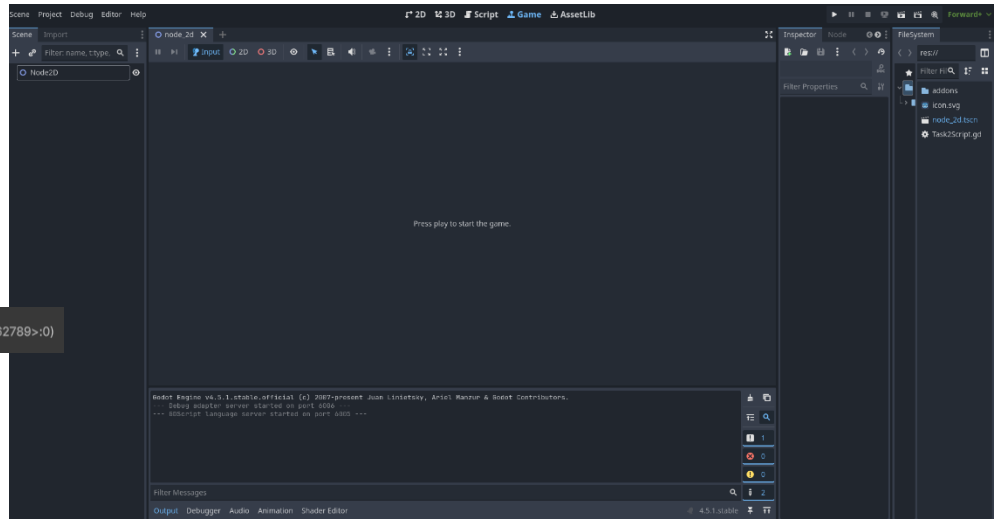
| Tasks | Unity | Godot |
|---|---|---|
| **TASK 7** | Change the Layout | Change the Layout |
| **Efficient** | The task was completed | The task was completed |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Effective** | I was able to complete this task with drags and drops | I was able to complete the task in 2 clicks |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Engaging** | Drag and drop is hard to make bad | Being able to put the window into the exact location I want is pretty engaging |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Error Tolerant** | The error messages for layouts were a little unwieldy | There was no direction on how to put windows back into the editor from floating |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Easy to Learn** | Drag ~~and Drop~~ | Placing the window was simple and straight forward |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / **Agree** 5 / **Strongly Agree** 6 |
| **Additional Notes** | To complete this task all I had to do was drag and drop the various windows.<br>Should you try to reset the layout clicking “Window”->”Layout”->”Reset All Layouts”, a very unwieldy error comes up. I point this out as a new time programmer may think this is the way to reset the layout, and may be left confused by the error message. | To change the layout, I had to click the vertical “…” drop down next to a window, then click the position I wanted the window in. You can also pop windows out (“Make Floating”).<br>If you pop a window out, it is not clear how to pop it back into a place (for the record, you close the window to place it back) |





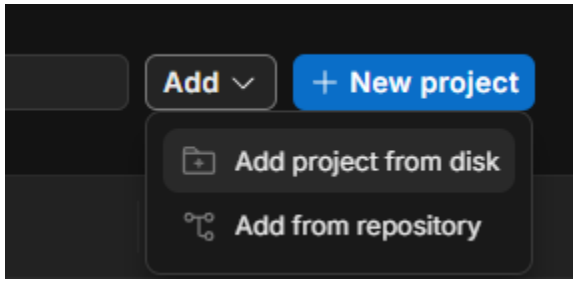



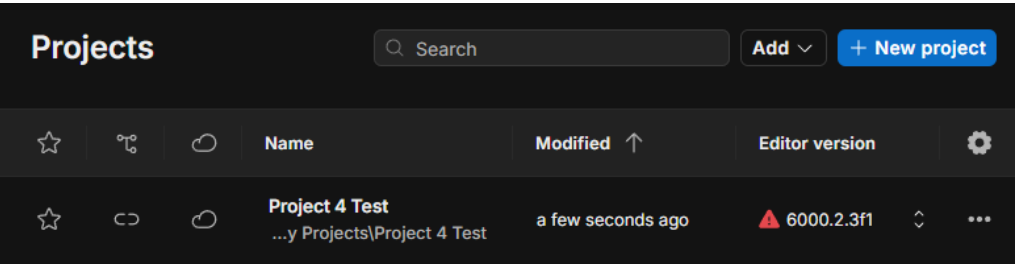
[02:08:38] IOException: Access to the path '\\?\C:\Users\conno_e7e4dwm\OneDrive\MyProjects\Unity Projects\Project 4 Test\UserSettings\Layouts' is denied.
System.IO.FileSystem.RemoveDirectoryInternal (System.String fullPath, System.Boolean topLevel, System.Boolean allowDirectoryNotEmpty) (at <59bd7c40c082431db25e1e728ab62789>:0)

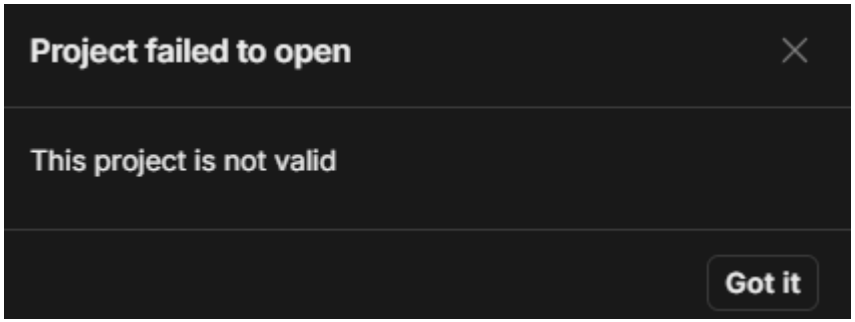


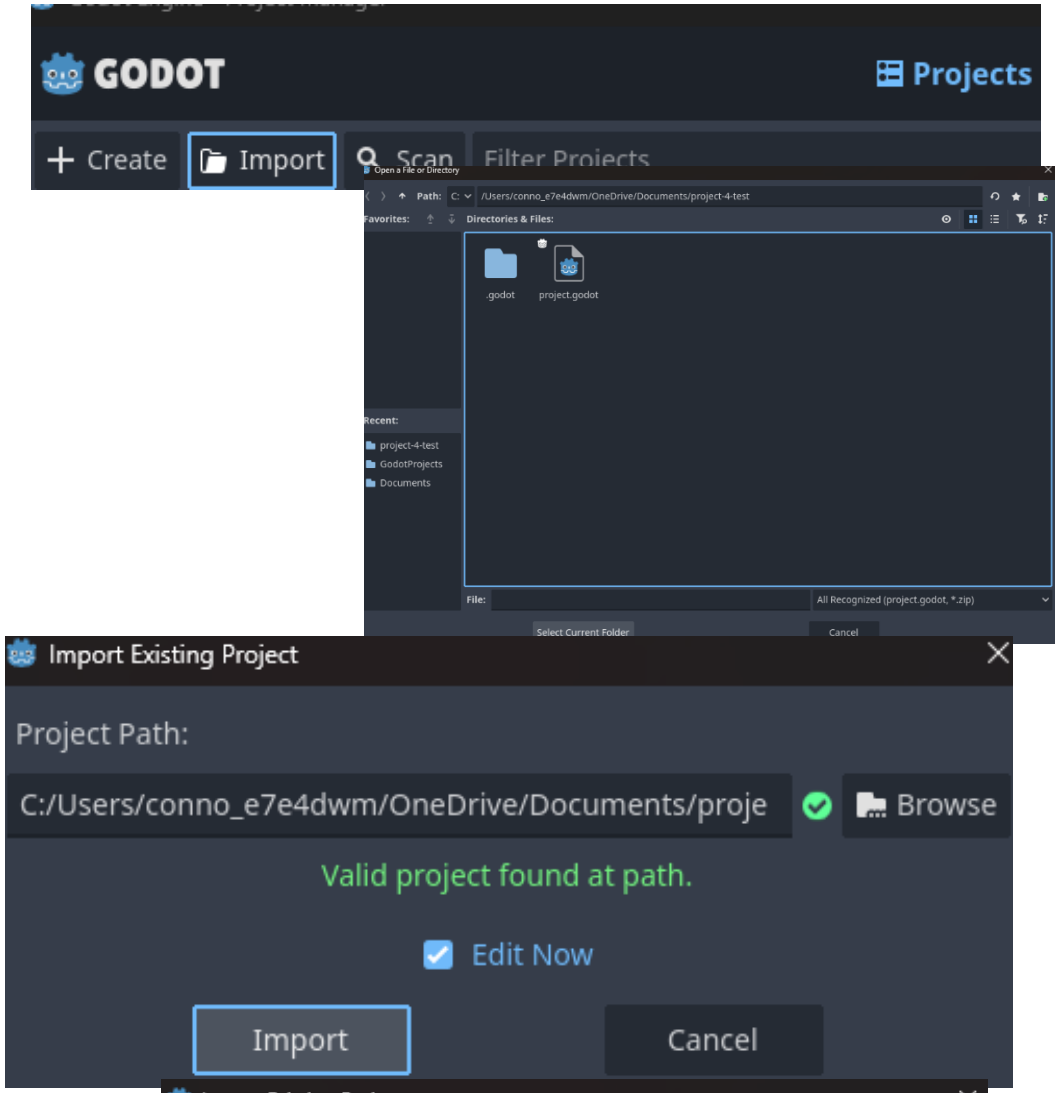
## 6.8 | Import an Existing Project

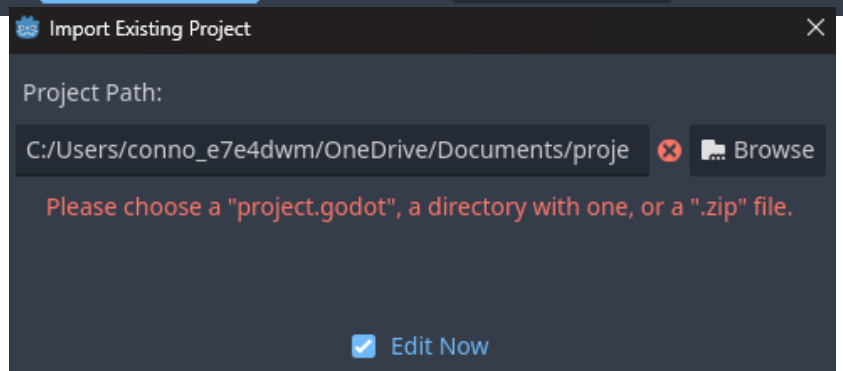
| Tasks | Unity | Godot |
|---|---|---|
| **TASK 8** | Import an Existing Project | Import an Existing Project |
| **Efficient** | I was able to complete the task | I was able to complete the task |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / Agree 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / Agree 5 / **Strongly Agree** 6 |
| **Effective** | The task was completed in 3 clicks, with some light file navigation | The task was completed in 3 clicks, with some light file navigation |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / Agree 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / Agree 5 / **Strongly Agree** 6 |
| **Engaging** | Unity does the standard thing for most software and uses the system's local file explorer for file navigation | The windows and folder navigation being done within the godot software was engaging |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / Agree 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / Agree 5 / **Strongly Agree** 6 |
| **Error Tolerant** | The error message(s) for this task are not very specific on feedback for why importing the project doesn't work | Godot provided helpful feedback in its error message |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / Agree 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / Agree 5 / **Strongly Agree** 6 |
| **Easy to Learn** | Import worked just as it should, and so was intuitive | Import worked just as it should, and so was very intuitive |
| | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / Agree 5 / **Strongly Agree** 6 | **Strongly Disagree** 1 / Disagree 2 / **Slightly Disagree** 3 / **Slightly Agree** 4 / Agree 5 / **Strongly Agree** 6 |
| **Additional Notes** | To import a project I had to click "Add"->"Add project from disk". Then locate the folder that has the name of my project, open that folder, then click "Open"". And the project is added to my list of projects.<br>Should I try to open a folder without a valid project, an error pops up that says "This project is not valid" which is not very specific on why | To import the project I first clicked "Import" on the Godot main screen, located the project in the new window, clicked "Select Current Folder", then Import and the project opened.<br>Should I try to open a folder without a valid project, a helpful error pops up with instruction and how to open a valid project. |


Add
+ New project
Add project from disk
Add from repository


Projects
Search
Add
+ New project
Name
Modified
Editor version
Project 4 Test
...y Projects\Project 4 Test
a few seconds ago
6000.2.3f1

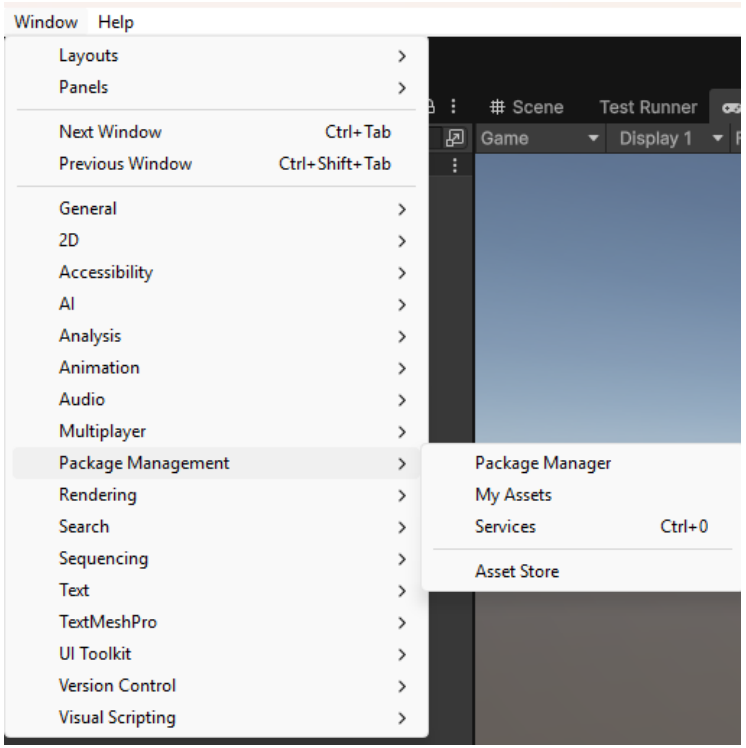

Project failed to open
This project is not valid
Got it

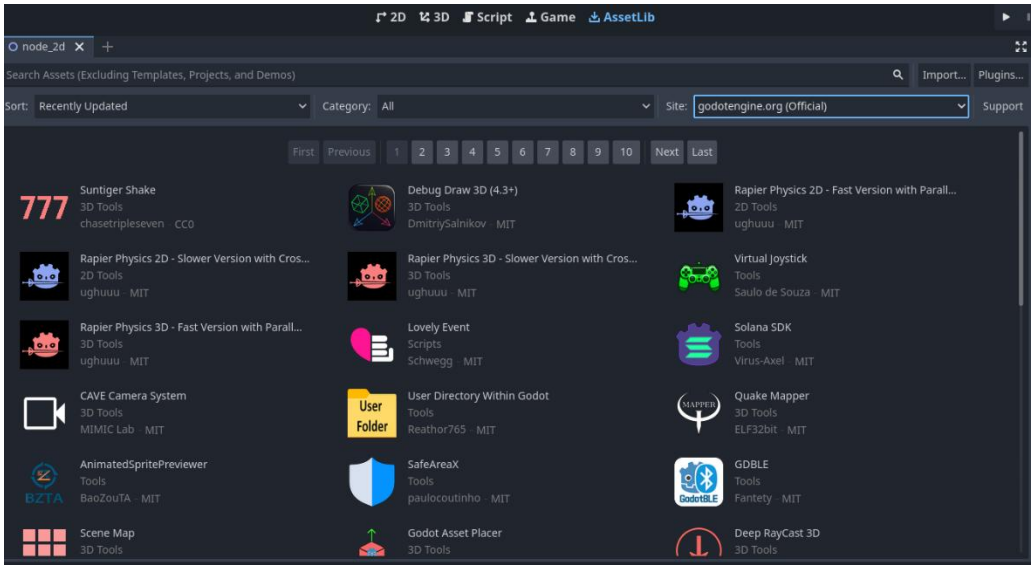

GODOT
Projects
+ Create
Import
Scan
Filter Projects
Import Existing Project
Project Path:
C:/Users/conno_e7e4dwm/OneDrive/Documents/proje
Browse
Valid project found at path.
Edit Now
Import
Cancel

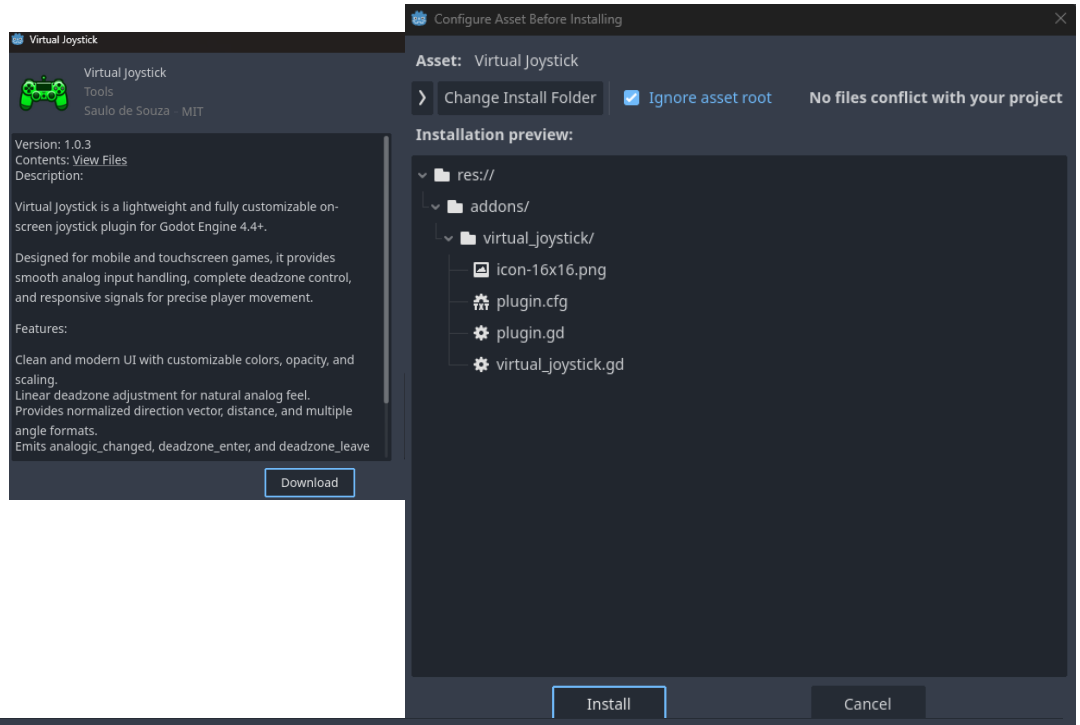

Import Existing Project
Project Path:
C:/Users/conno_e7e4dwm/OneDrive/Documents/proje
Browse
Please choose a "project.godot", a directory with one, or a ".zip" file.
Edit Now

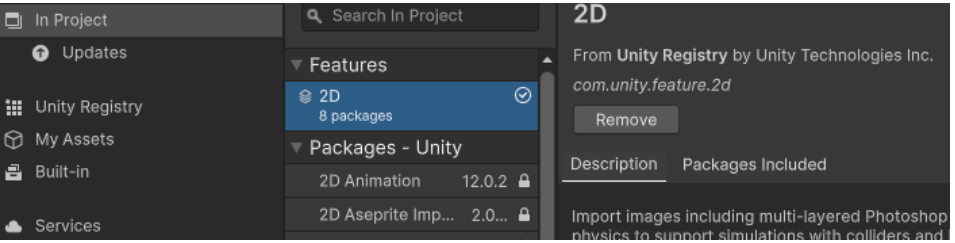
## 6.9 | Install an Extension

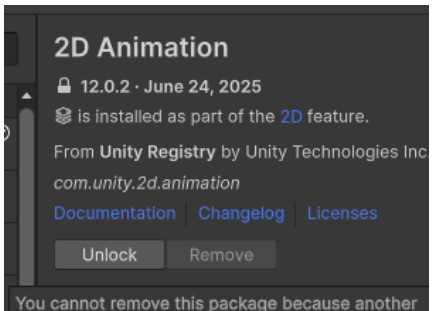
| Tasks | Unity | Godot |
|---|---|---|
| **TASK 9** | Install an extension | Install an extension |
| **Efficient** | The task was completed | The task was completed |
| | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 |
| **Effective** | The task was completed in 6 clicks | It took 4 clicks to complete this task |
| | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 |
| **Engaging** | The package manager is a bit plain, but effective | Getting the extension was very engaging |
| | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 |
| **Error Tolerant** | The error message explains the why and how you can't remove sections of an extension | Error message was clear and direct |
| | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 |
| **Easy to Learn** | Finding the package manager is not intuitive. | Getting the extension was straight forward and simple |
| | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 |
| **Additional Notes** | Unity does not natively support external (there is a Unity Website with assets and extensions, but most are paid assets/tools) extensions, so to complete this task I imported a package, which are Unity native ('official') extensions. I clicked "Window"->"Package Management"->"Package Manager". Then clicked "Unity Registry"->"2D" (to import the 2D unity extension/package)->"Install. To get my extension Should I try to remove a section of the extension, Unity will display a helpful message to explain why I can't | To install an extension I went to the AssetLib tab and selected "go online" (I clicked on it before screenshotting). Then was given many choices for extensions to download. I selected "Virtual Joystick" from MIT. Clicked Download and then install on the next page to receive my extension |

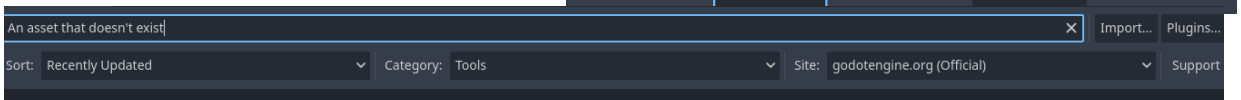

Window
Help
Layouts
Panels
Next Window Ctrl+Tab
Previous Window Ctrl+Shift+Tab
General
2D
Accessibility
AI
Analysis
Animation
Audio
Multiplayer
Package Management
Rendering
Search
Sequencing
Text
TextMeshPro
UI Toolkit
Version Control
Visual Scripting
Scene
Test Runner
Game
Display 1
Package Manager
My Assets
Services Ctrl+0
Asset Store


2D
3D
Script
Game
AssetLib
node_2d
Search Assets (Excluding Templates, Projects, and Demos)
Import...
Plugins...
Sort: Recently Updated
Category: All
Site: godotengine.org (Official)
Support
First
Previous
Next
Last
Suntiger Shake
3D Tools
chasetripleseven · CC0
Debug Draw 3D (4.3+)
3D Tools
DmitriySalnikov · MIT
Rapier Physics 2D - Fast Version with Parall...
2D Tools
ughuuu · MIT
Rapier Physics 2D - Slower Version with Cros...
2D Tools
ughuuu · MIT
Rapier Physics 3D - Slower Version with Cros...
3D Tools
ughuuu · MIT
Virtual Joystick
Tools
Saulo de Souza · MIT
Rapier Physics 3D - Fast Version with Parall...
3D Tools
ughuuu · MIT
Lovely Event
Scripts
Schwegg · MIT
Solana SDK
Tools
Virus-Axel · MIT
CAVE Camera System
3D Tools
MIMIC Lab · MIT
User Directory Within Godot
Tools
Reathor765 · MIT
Quake Mapper
3D Tools
ELF32bit · MIT
AnimatedSpritePreviewer
Tools
BaoZouTA · MIT
SafeAreaX
Tools
paulocoutinho · MIT
GDBLE
Tools
Fantety · MIT
Scene Map
3D Tools
Godot Asset Placer
3D Tools
Deep RayCast 3D
3D Tools


Virtual Joystick
Virtual Joystick
Tools
Saulo de Souza · MIT
Version: 1.0.3
Contents: View Files
Description:
Virtual Joystick is a lightweight and fully customizable on-screen joystick plugin for Godot Engine 4.4+.
Designed for mobile and touchscreen games, it provides smooth analog input handling, complete deadzone control, and responsive signals for precise player movement.
Features:
Clean and modern UI with customizable colors, opacity, and scaling.
Linear deadzone adjustment for natural analog feel.
Provides normalized direction vector, distance, and multiple angle formats.
Emits analogic_changed, deadzone_enter, and deadzone_leave
Download
Configure Asset Before Installing
Asset: Virtual Joystick
Change Install Folder
Ignore asset root
No files conflict with your project
Installation preview:
res://
addons/
virtual_joystick/
icon-16x16.png
plugin.cfg
plugin.gd
virtual_joystick.gd
Install
Cancel

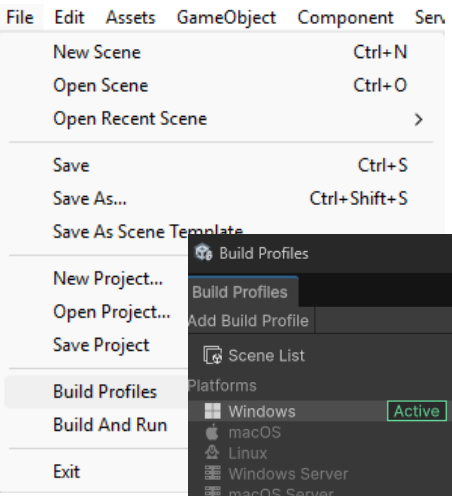

In Project
Updates
Unity Registry
My Assets
Built-in
Services
Search In Project
Features
2D
8 packages
Packages - Unity
2D Animation 12.0.2
2D Aseprite Imp... 2.0...
2D
From Unity Registry by Unity Technologies Inc.
com.unity.feature.2d
Remove
Description
Packages Included
Import images including multi-layered Photoshop

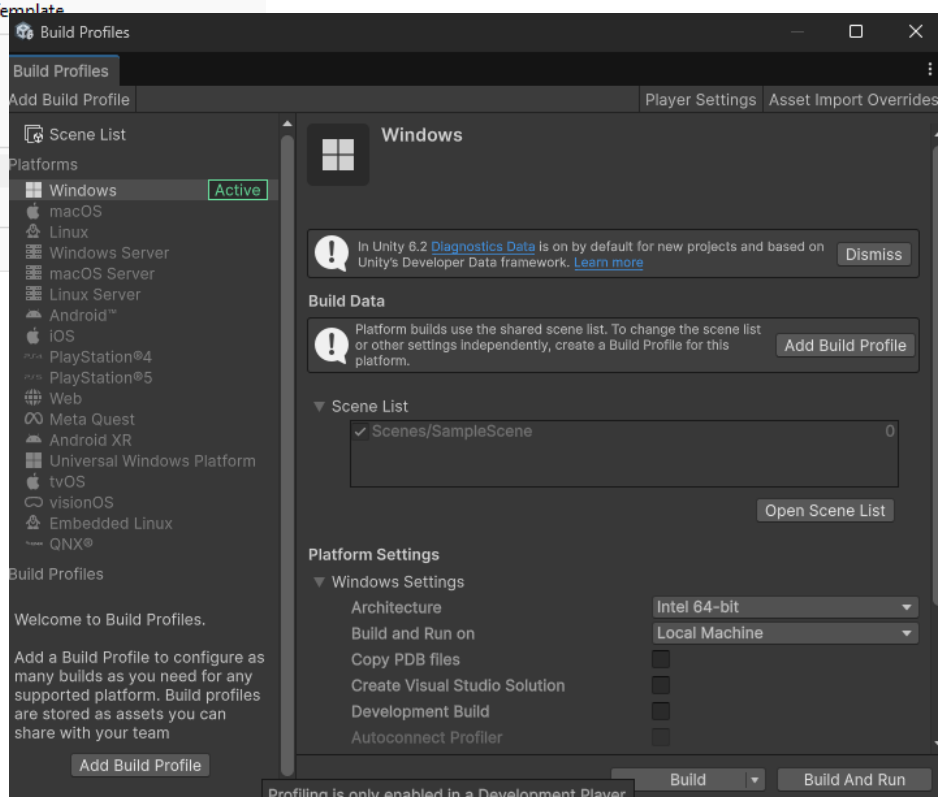

2D Animation
12.0.2 · June 24, 2025
is installed as part of the 2D feature.
From Unity Registry by Unity Technologies Inc
com.unity.2d.animation
Documentation
Changelog
Licenses
Unlock
Remove
You cannot remove this package because another

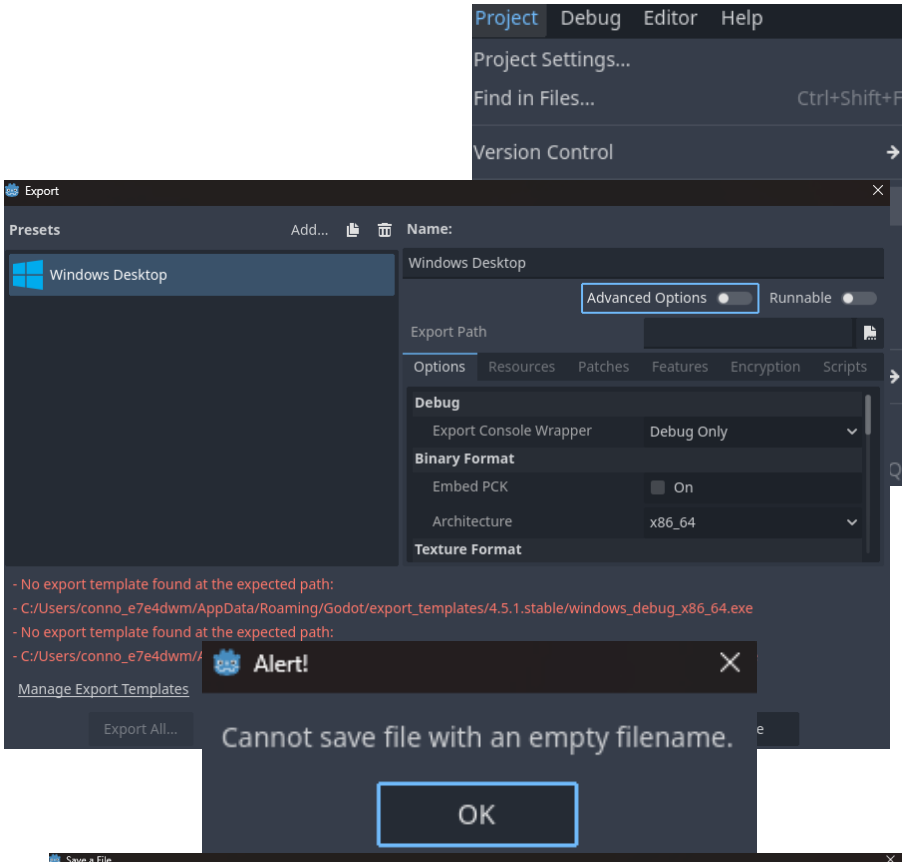

An asset that doesn't exist
Import...
Plugins...
Sort: Recently Updated
Category: Tools
Site: godotengine.org (Official)
Support

## 6.10 | Export/Build the Project

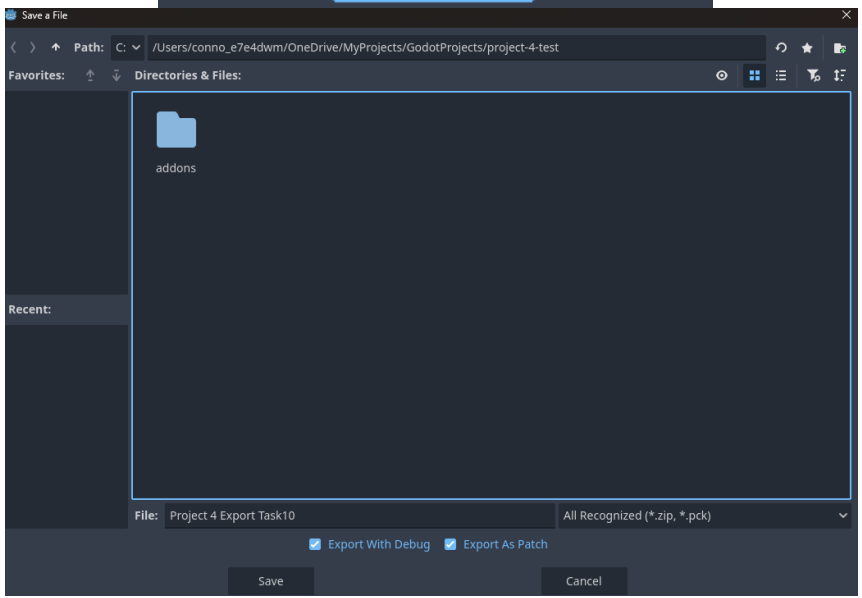
| Tasks | Unity | Godot |
|---|---|---|
| **TASK 10** | Export/Build the Project | Export/Build the Project |
| **Efficient** | The task was completed | The task was completed |
| | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 |
| **Effective** | The task was completed in 3 clicks | The task was completed in 3 clicks and writing a file name |
| | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 |
| **Engaging** | The Unity UI is relatively straightforward for this task, though "Build profiles" doesn't exactly scream "Build here" | Godot gives simple instructions on how to save the project |
| | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 |
| **Error Tolerant** | Unity's error here is very clear and helpful, and provides clear direction on how to build properly | Godot has many useful error messages for this task |
| | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 |
| **Easy to Learn** | The Unity UI is pretty straightforward on what to do to build your project, but understanding how to set build setting up properly may take some learning | Exporting takes some setup and is a little confusing, but not too bad. |
| | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 | **Strongly Disagree** 1 / Disagree 2 / Slightly Disagree 3 / Slightly Agree 4 / Agree 5 / Strongly Agree 6 |
| **Additional Notes** | To export the project, I clicked "File"->"Build Profiles". Then in the Build Profiles window I clicked build (Unity assumes your building for the same type of system, in this case Windows, as the editor currently running on), and after selecting a file to put the build in and some mild loading it completes a build of the project.<br>Should you select to build in the same folder as the project, Unity provides a popup to explain why the error occurred, and what to do next to fix it. | To export the project I went to the Project tab and clicked "Export…", after setting the build settings, I clicked export, then save on the next window. Then the project closed out without leaving any instructions on what to do next or what it did exactly.<br>If you did not set the project up properly for your build settings, Godot has some lengthy error messages to inform you.<br>If you do not create a file name, Godot informs you that you need a file name for the export. |

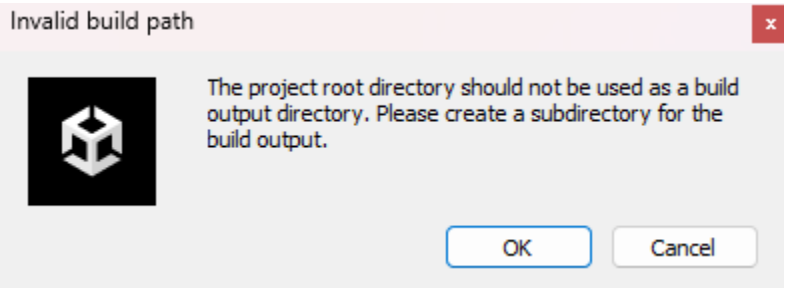

File Edit Assets GameObject Component Ser
New Scene Ctrl+N
Open Scene Ctrl+O
Open Recent Scene
Save Ctrl+S
Save As... Ctrl+Shift+S
Save As Scene Template
New Project...
Open Project...
Save Project
Build Profiles
Build And Run
Exit


Build Profiles
Build Profiles
Add Build Profile
Player Settings
Asset Import Overrides
Scene List
Platforms
Windows
Active
macOS
Linux
Windows Server
macOS Server
Linux Server
Android™
iOS
PlayStation®4
PlayStation®5
Web
Meta Quest
Android XR
Universal Windows Platform
tvOS
visionOS
Embedded Linux
QNX®
Build Profiles
Welcome to Build Profiles.
Add a Build Profile to configure as many builds as you need for any supported platform. Build profiles are stored as assets you can share with your team
Add Build Profile
Windows
In Unity 6.2 Diagnostics Data is on by default for new projects and based on Unity's Developer Data framework. Learn more
Dismiss
Build Data
Platform builds use the shared scene list. To change the scene list or other settings independently, create a Build Profile for this platform.
Add Build Profile
Scene List
Scenes/SampleScene
0
Open Scene List
Platform Settings
Windows Settings
Architecture
Intel 64-bit
Build and Run on
Local Machine
Copy PDB files
Create Visual Studio Solution
Development Build
Autoconnect Profiler
Build
Build And Run


Project Debug Editor Help
Project Settings...
Find in Files... Ctrl+Shift+F
Version Control
Export
Presets
Add...
Windows Desktop
Name:
Windows Desktop
Advanced Options
Runnable
Export Path
Options Resources Patches Features Encryption Scripts
Debug
Export Console Wrapper
Debug Only
Binary Format
Embed PCK
On
Architecture
x86_64
Texture Format
- No export template found at the expected path:
- C:/Users/conno_e7e4dwm/AppData/Roaming/Godot/export_templates/4.5.1.stable/windows_debug_x86_64.exe
- No export template found at the expected path:
- C:/Users/conno_e7e4dwm/
Manage Export Templates
Export All...


Alert!
Cannot save file with an empty filename.
OK


Save a File
Path:
C:
/Users/conno_e7e4dwm/OneDrive/MyProjects/GodotProjects/project-4-test
Favorites:
Directories & Files:
addons
Recent:
File:
Project 4 Export Task10
All Recognized (*.zip, *.pck)
Export With Debug
Export As Patch
Save
Cancel


Invalid build path
The project root directory should not be used as a build output directory. Please create a subdirectory for the build output.
OK
Cancel